



ENGR 689 - Multimodal LLM Agents

Day 1: Large Language Models

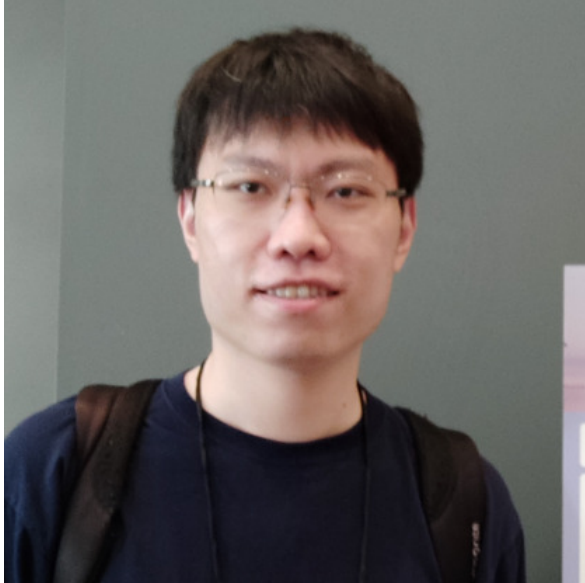
Yu Zhang

yuzhang@tamu.edu

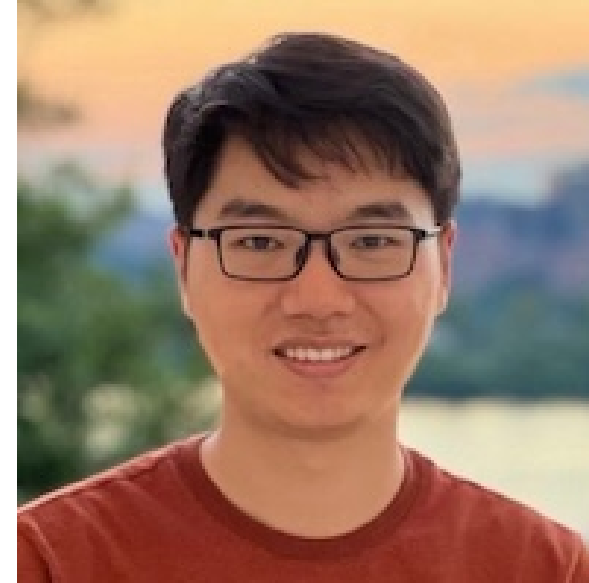
(Co-taught with Prof. Cheng Zhang)

Instructors

- Yu Zhang (Assistant Professor, CSE)
- Email: yuzhang@tamu.edu
- Office: PETR 222
- Office Hour: 2-3 PM Tuesday & Thursday



- Cheng Zhang (Assistant Professor, CSE)
- Email: chzhang@tamu.edu
- Office: PETR 321
- Office Hour: 2-3 PM Tuesday & Thursday



Grading

- Quizzes: 40%
 - Quiz 1: 10% [during the Day 1 class (i.e., today)]
 - Quiz 2: 10% [during the Day 2 class]
 - Quiz 3: 10% [during the Day 3 class]
 - Quiz 4: 10% [during the Day 4 class]
- Group Project: 60%
 - Presentation: 20% [during the Day 5 class]
 - Slides: 20% [due 11:59 PM, Day 5]
 - Code/Demo: 20% [due 11:59 PM, Day 5]

4 Quizzes (10% × 4 = 40%)

- In-class
- 25 minutes
- 8-10 multiple-choice questions
 - Most questions are worth 1%, with a few worth 2%
- Open-book
 - You may use your laptop to access ANY resources, but you need to manage your time wisely
- The questions will cover material from that day's **Foundations and Core Concepts** section
- **Motivation**: attendance check + making sure you follow the lecture and know roughly where to find the relevant material

Daily Schedule for Day 1 – Day 4

**Foundations and
Principles**
(90 minutes)

Break
(30 minutes)

Quiz
(25 minutes)

Practical Tools
(60 minutes)

Group Project (60%)

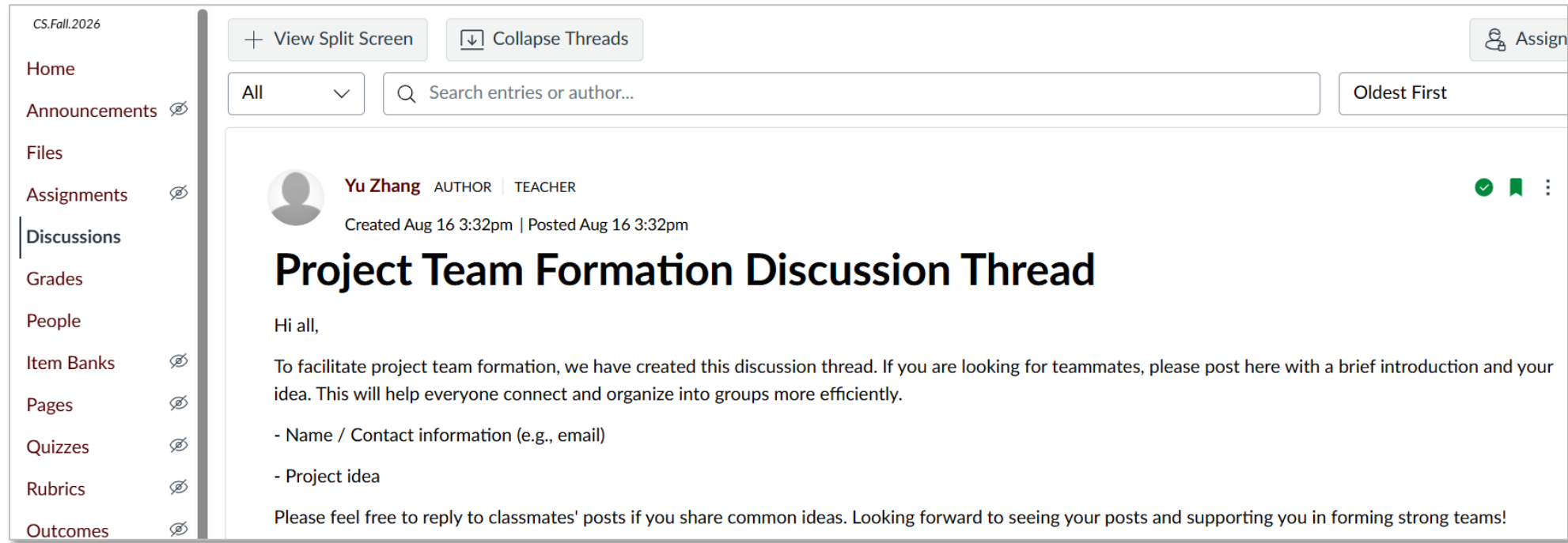
- Teams of **2 or 3**
 - Any deviation from this size requires prior approval from the instructors
 - If you need to request an exception, please email the instructor(s) by **11:59 PM on Day 2**
 - If too many people request to work individually, we may merge some of you into groups.
- **Goal:** Build a practical, demo-friendly LLM agent
 - **Example 1:** a web agent for booking tickets
 - **Example 2:** a research agent that turns a research paper into a website
 - You may use any open/closed-source LLM as the backbone
 - Your agent may be **multimodal** or **primarily text-based**
 - You may, and **are encouraged to**, use any **vibe-coding tools** (e.g., Codex, which we will cover in this course) to help you build quickly

Group Project (60%)

- **Goal:** Build a practical, demo-friendly LLM agent
- **Presentation (20%):** Each team will give a 15-minute presentation on Day 5, including a live or video demo showing how the agent works.
- **Slides (20%):** Each team should upload the slides to Canvas by 11:59 PM on Day 5
- **Code/Demo (20%):** Each team should upload the code link and demo link to Canvas by 11:59 PM on Day 5
 - The demo may be a recorded video, a live demo hosted on Hugging Face, or a GitHub webpage (e.g., <https://agentflow.stanford.edu>)
- Detailed rubrics for each component will be posted as Canvas announcements

Group Project (60%)

- If you are unsure **whom to work with**, please use the team-finding post on Canvas.



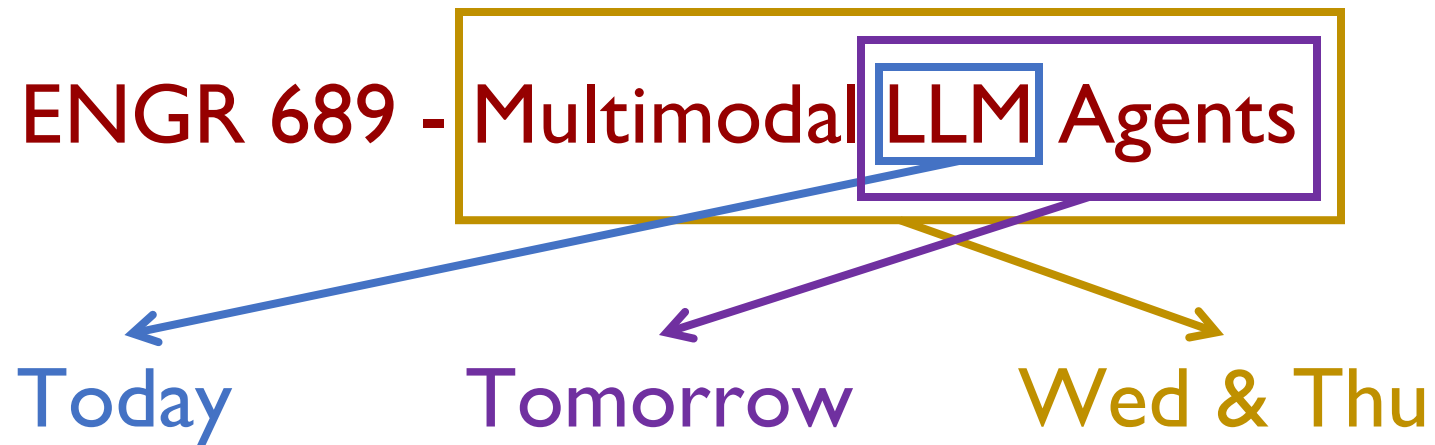
The screenshot shows a Canvas LMS interface for a course titled "CS.Fall.2026". On the left is a navigation sidebar with links to Home, Announcements, Files, Assignments, Discussions (which is highlighted), Grades, People, Item Banks, Pages, Quizzes, Rubrics, and Outcomes. The main content area displays a discussion thread. At the top of the thread are buttons for "View Split Screen" and "Collapse Threads", along with an "Assign" button. Below these is a search bar and a dropdown menu set to "All". The thread is titled "Project Team Formation Discussion Thread" and is authored by "Yu Zhang", who is identified as an "AUTHOR" and "TEACHER". The thread was created and posted on "Aug 16 3:32pm". The content of the thread begins with "Hi all," followed by a paragraph explaining the purpose of the thread: "To facilitate project team formation, we have created this discussion thread. If you are looking for teammates, please post here with a brief introduction and your idea. This will help everyone connect and organize into groups more efficiently." It then lists two requirements for posts: "- Name / Contact information (e.g., email)" and "- Project idea". The thread concludes with the statement: "Please feel free to reply to classmates' posts if you share common ideas. Looking forward to seeing your posts and supporting you in forming strong teams!".

- If you are unsure **what topic to work on**, you are very welcome to come to office hours and discuss ideas with either instructor.

Other Course Logistics

- **Absence Policy:** Please refer to Student Rule 7 (<https://student-rules.tamu.edu/rule07/>) about **excused absences**, including definitions, and related documentation and timelines.
 - Please email the instructors if you believe your situation qualifies as an excused absence. We will work with you to make appropriate arrangements for the quiz and/or final presentation.
- **Regrade Policy:** Grading for multiple-choice questions is usually straightforward, but if you believe there is an issue with the grading, please email us.
- **AI Policy:** Use AI freely and **extensively**
 - A key goal of this sprint course is to help you learn how to use AI tools to rapidly build projects that would otherwise be very time-consuming to develop from scratch.

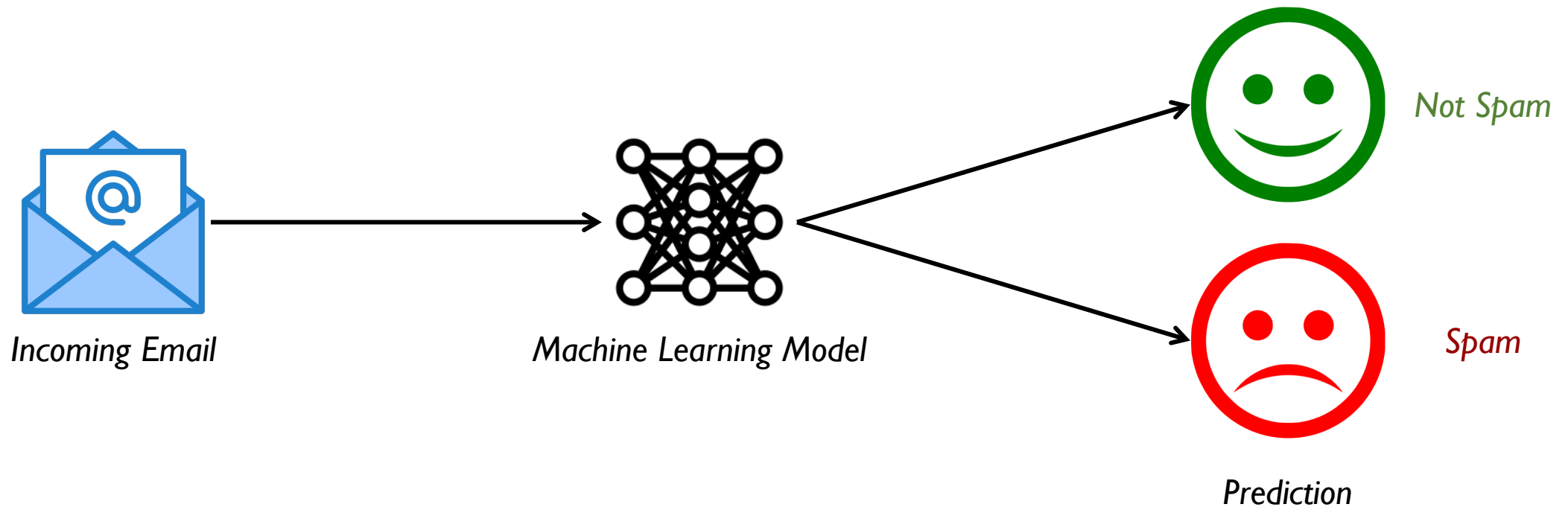
What does this course cover?



Let's start with the language modality.

Background: (Supervised) Machine Learning

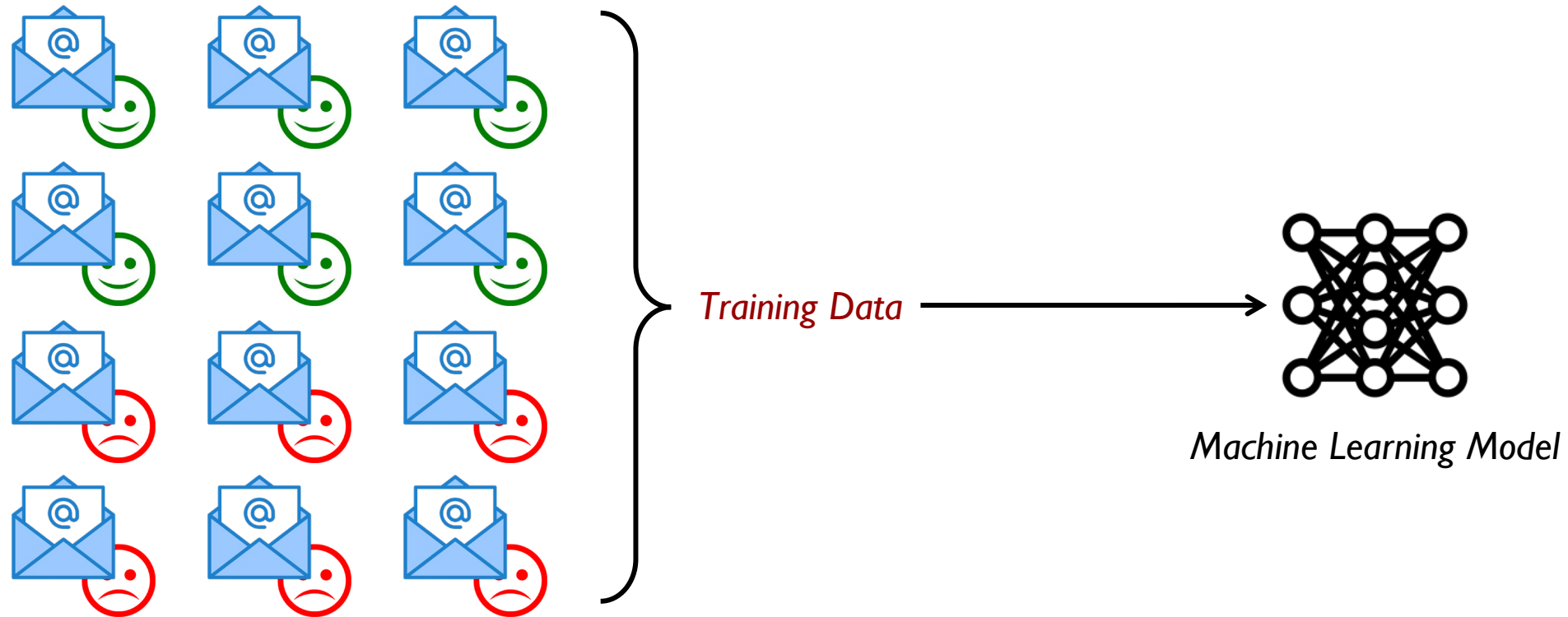
- Suppose we want to build a spam email detector.
 - Our desired outcome:



- But how do we get this model?

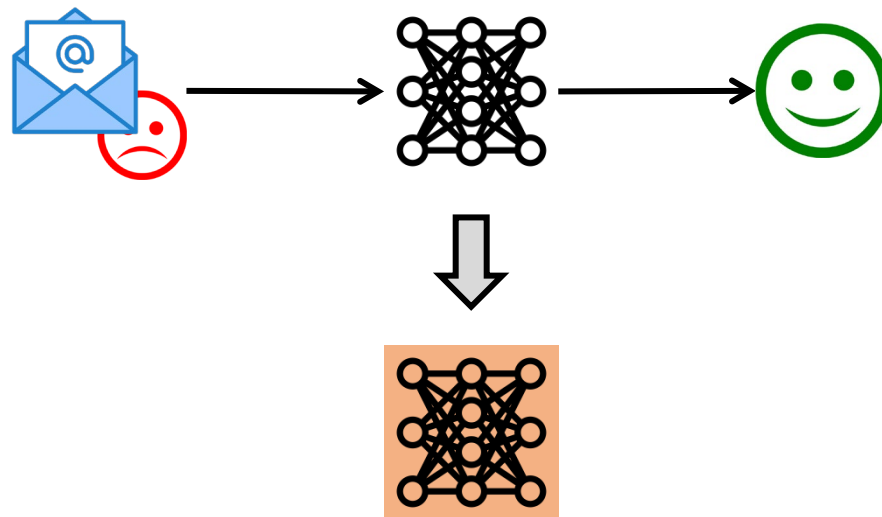
Background: (Supervised) Machine Learning

- But how do we get this model?
- (Supervised) Training:
 - We collect many emails and have people label each one as spam or not spam.



Background: (Supervised) Machine Learning

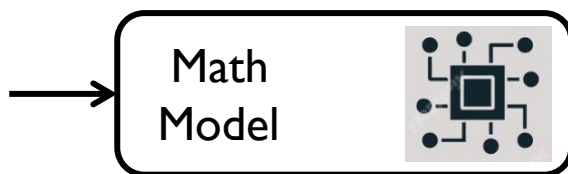
- But how do we get this model?
- (Supervised) Training:
 - Given a training email, we **hide its label** and ask the model to predict it.
 - If the prediction is wrong, the model still needs improvement.
 - We update the model (parameters) using an optimization algorithm.



Natural Language Processing before the Era of LLMs

- Given a natural language processing (NLP) task
 - Step 1:** Find/annotate **task-specific** training data (e.g., 10,000 training samples)
 - Step 2:** Design a **task-specific** model (e.g., 3-layer recurrent neural network + classification head)
 - Step 3:** Use the annotated data to train the model

Math Word Problem:
(John buys 20 cards and 1/4 are uncommon. How many uncommon cards did he get?, 5)



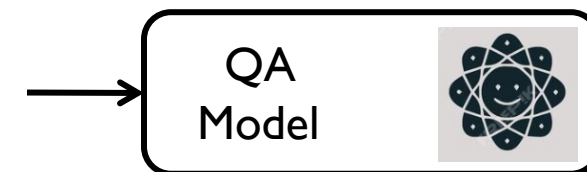
Paper Classification: (Rapid chromatographic technique for preparative separations with moderate resolution, Organic Chemistry)



Information Extraction: (in rats, nitrofurantoin causes pulmonary toxicity, [pulmonary toxicity, DISEASE])

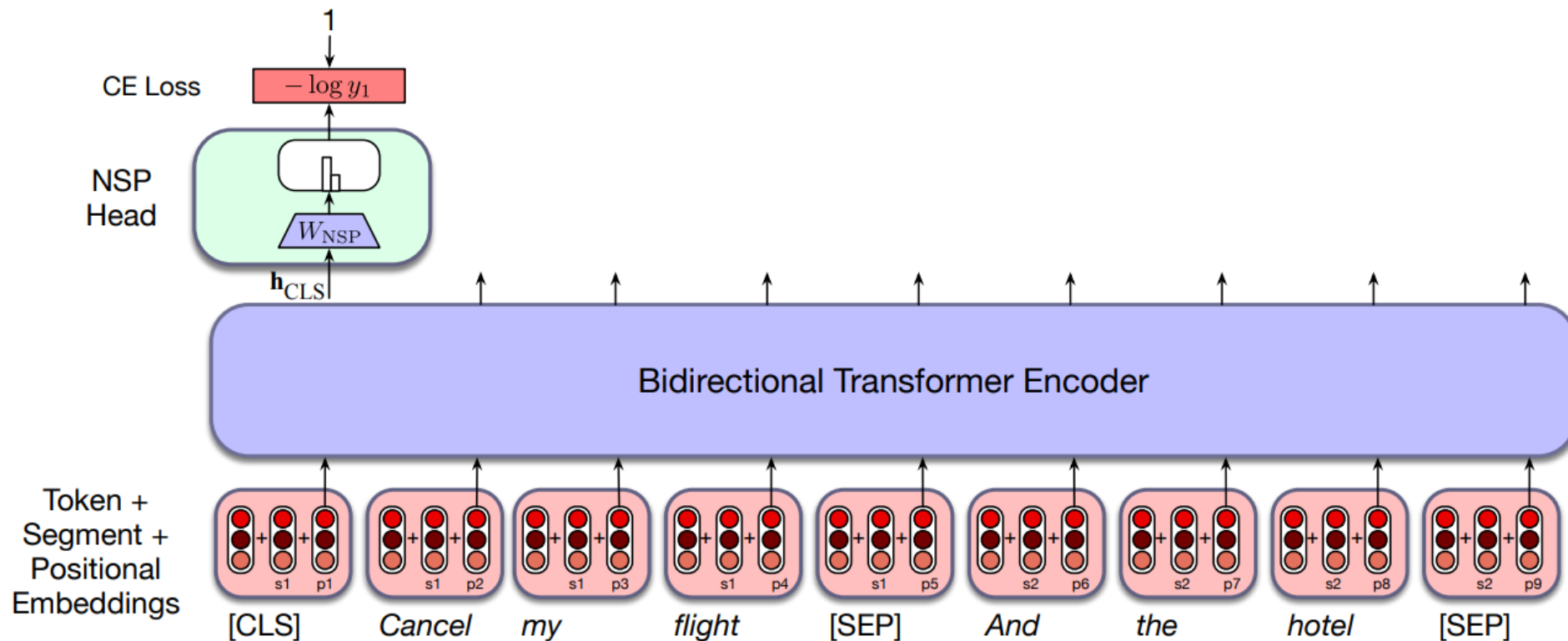


Question Answering: (Who formulated the zeroth law of thermodynamics?, Ralph H. Fowler)

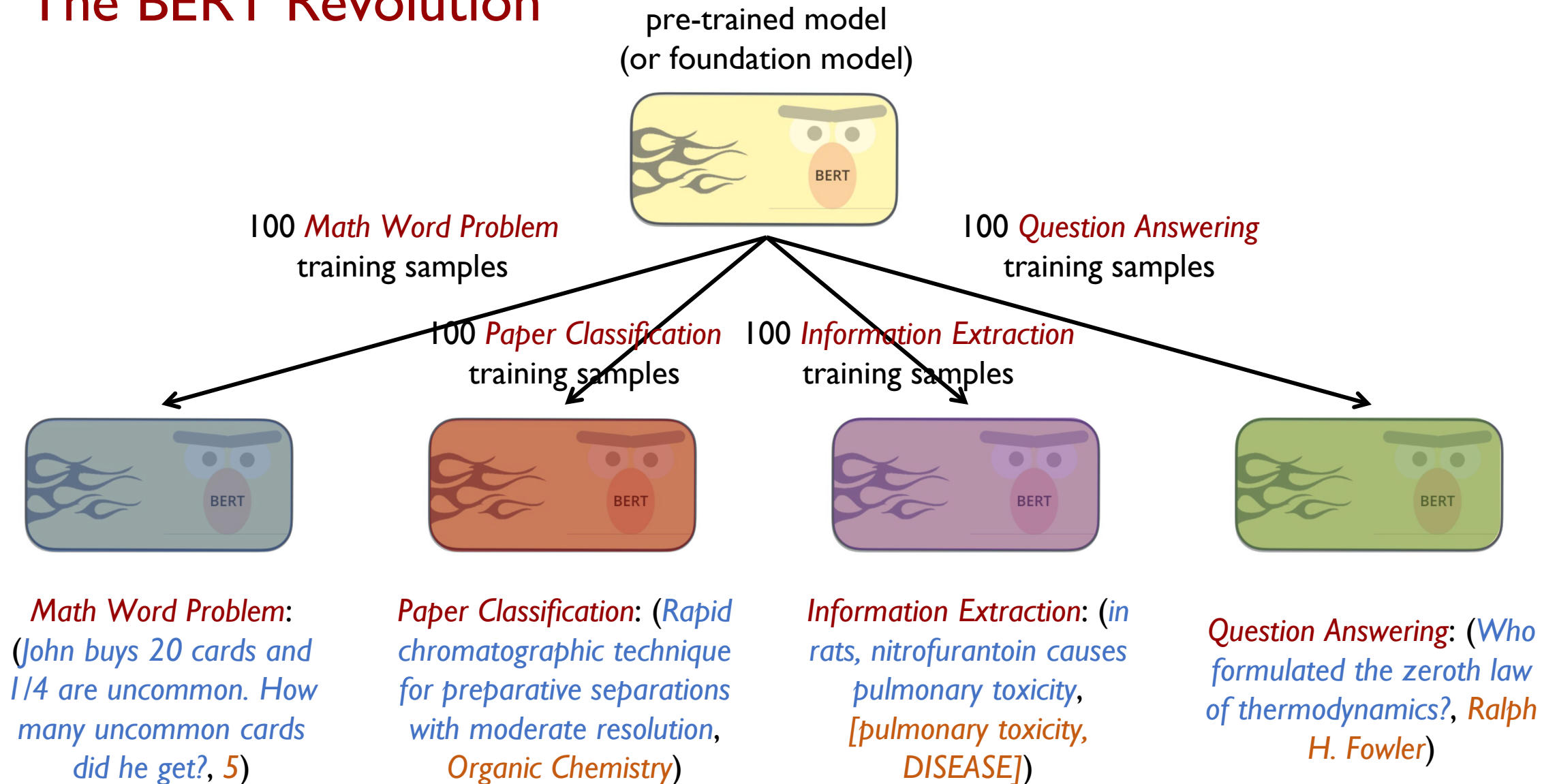


A Breakthrough: BERT [Devlin et al., 2019]

- A **Transformer** architecture with 12 or 24 layers
- **Pre-trained** on massive text corpora (the entire Wikipedia + BookCorpus)
- Learns **linguistic patterns** and **general knowledge** useful across a wide range of NLP tasks



The BERT Revolution



But today's ChatGPT is far more powerful!

Solve the following Math Word Problem:

John buys 20 cards and 1/4 are uncommon. How many uncommon cards did he get?

0 Math Word Problem training samples

Classify the following paper. Candidate classes are [XXX, XXX, ...]:
Rapid chromatographic technique for preparative separations with ...

0 Paper Classification training samples

Find all the DISEASE entities in the following sentence:
in rats, nitrofurantoin causes pulmonary toxicity

0 Information Extraction training samples

Answer the following question:
Who formulated the zeroth law of thermodynamics?

0 Question Answering training samples



One Model for All Tasks!

5

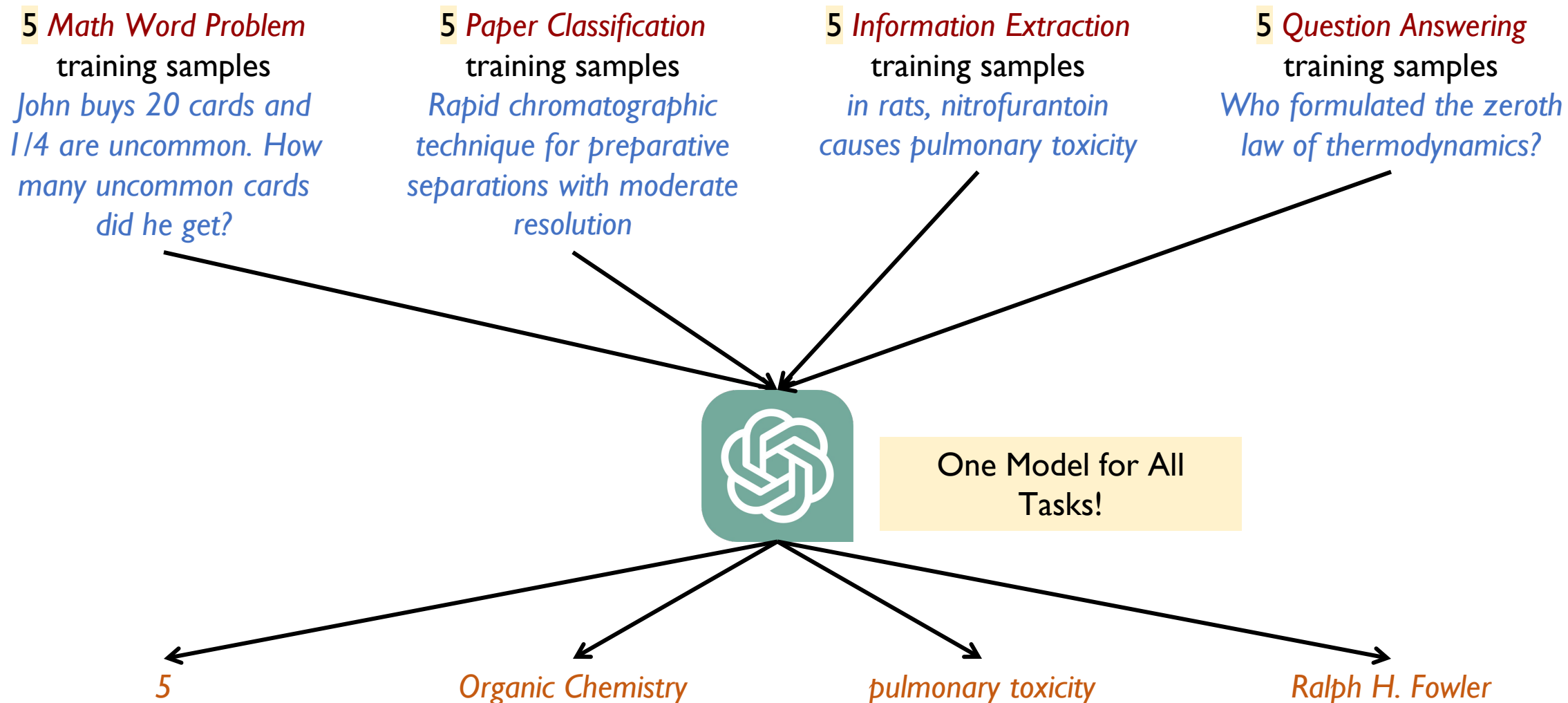
Organic Chemistry

pulmonary toxicity

Ralph H. Fowler

Let's take a step back first

The **task instruction** is optional if you have 5 training samples.



GPT-3 [Brown et al., 2020]

Language Models are Few-Shot Learners

Tom B. Brown*	Benjamin Mann*	Nick Ryder*	Melanie Subbiah*	
Jared Kaplan[†]	Prafulla Dhariwal	Arvind Neelakantan	Pranav Shyam	Girish Sastry
Amanda Askell	Sandhini Agarwal	Ariel Herbert-Voss	Gretchen Krueger	Tom Henighan
Rewon Child	Aditya Ramesh	Daniel M. Ziegler	Jeffrey Wu	Clemens Winter
Christopher Hesse	Mark Chen	Eric Sigler	Mateusz Litwin	Scott Gray
Benjamin Chess		Jack Clark	Christopher Berner	
Sam McCandlish	Alec Radford	Ilya Sutskever	Dario Amodei	

OpenAI

How was GPT-3 pre-trained?

- With only one task – next token prediction – on web-scale corpora

$$\sum_i p(\text{next token} \mid \text{previous tokens}; \Theta)$$

The diagram illustrates the next token prediction task. The equation shows the probability of a token w_i (labeled "next token" in an orange box) given the sequence of previous tokens $w_1, w_2, \dots, w_{i-1}$ (labeled "previous tokens" in a blue box), using the model parameters Θ (labeled "model parameters" in a green box).

- Most NLP tasks can “reduce” to next token prediction.
 - **Math:** {Input to the model} John buys 20 cards and 1/4 are uncommon. How many uncommon cards did he get? The answer is
 - {Model predicts the next token(s)} ||
 - **Classification:** {Input to the model} (paper title) Rapid chromatographic technique for preparative separations with moderate resolution => (label)
 - {Model predicts the next token(s)} Organic Chemistry

How was GPT-3 pre-trained?

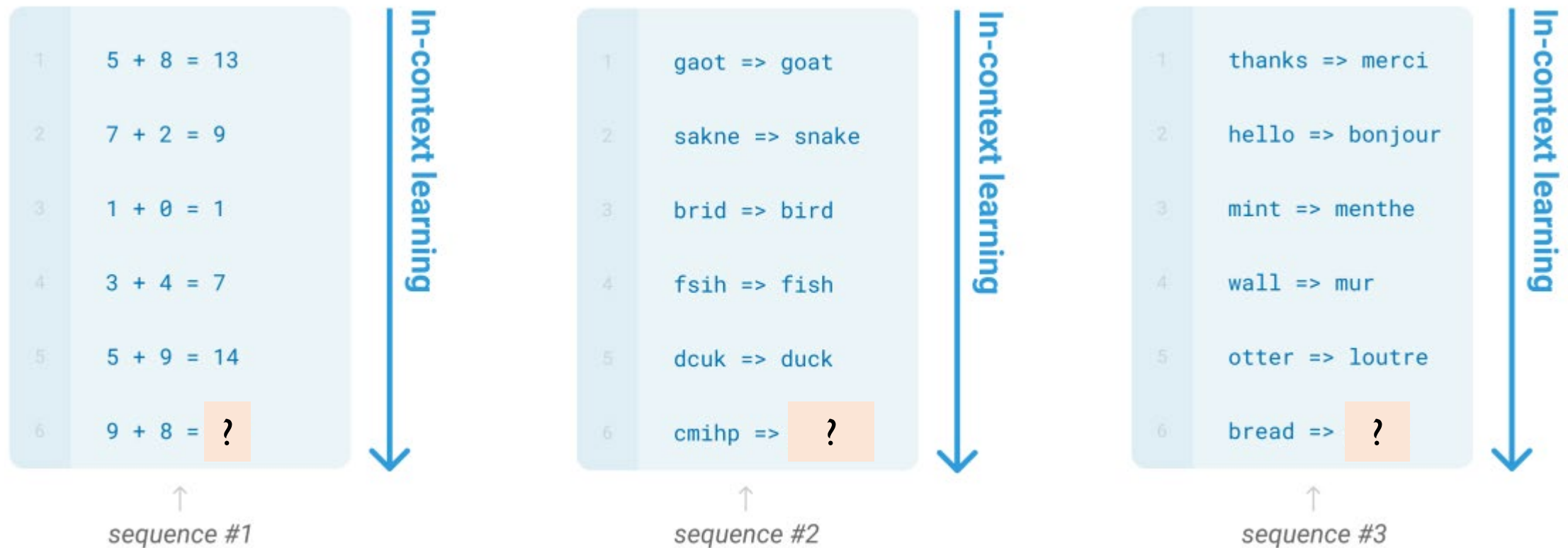
- With only one task – next token prediction – on web-scale corpora

$$\sum_i p(\underbrace{w_i}_{\text{next token}} \mid \underbrace{w_1, w_2, \dots, w_{i-1}}_{\text{previous tokens}}; \underbrace{\Theta}_{\text{model parameters}})$$

- Most NLP tasks can “reduce” to next token prediction.
 - **Information Extraction:** {Input to the model} (text) in rats, nitrofurantoin causes pulmonary toxicity. => (entity, type)
 - {Model predicts the next token(s)} pulmonary toxicity, disease
 - **Question Answering:** {Input to the model} Who formulated the zeroth law of thermodynamics?
 - {Model predicts the next token(s)} Ralph H. Fowler

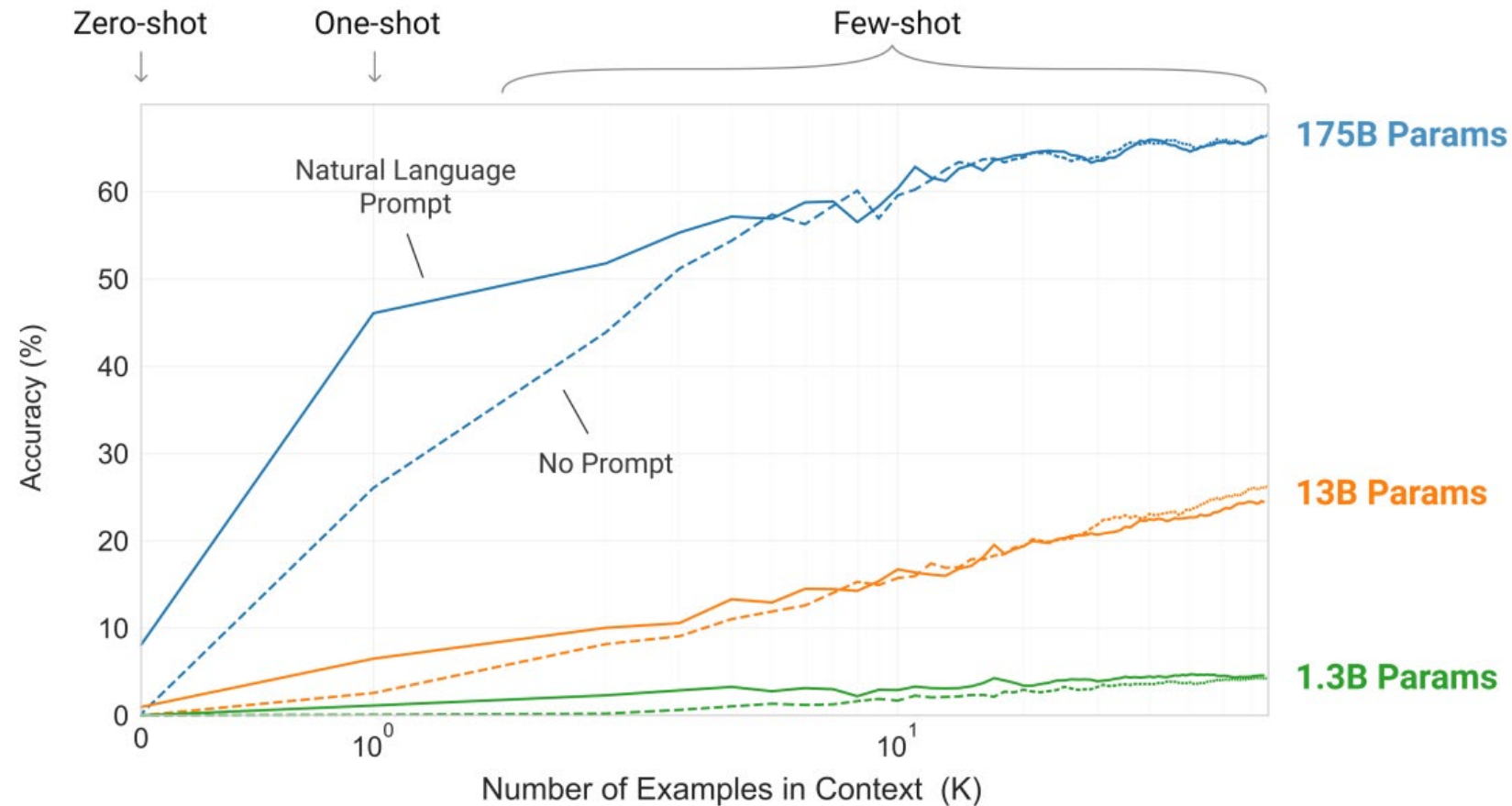
GPT-3 can perform each task with just a few examples

- The model may acquire a broad set of skills and pattern recognition abilities during pre-training. It then uses these abilities at inference time to rapidly adapt to or recognize the desired task. – “**In-context learning**”



Can a model be that “smart”?

- Only if it is big enough!



GPT-3

BERT-Base: 0.11B
BERT-Large: 0.34B

Can we make our demo examples more insightful?
(Prompt Engineering)

Chain-of-Thought Prompting [Wei et al., 2022]

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

Jason Wei Xuezhi Wang Dale Schuurmans Maarten Bosma
Brian Ichter Fei Xia Ed H. Chi Quoc V. Le Denny Zhou

Google Research, Brain Team
{jasonwei,dennyzhou}@google.com

Abstract

We explore how generating a *chain of thought*—a series of intermediate reasoning steps—significantly improves the ability of large language models to perform complex reasoning. In particular, we show how such reasoning abilities emerge naturally in sufficiently large language models via a simple method called *chain-of-thought prompting*, where a few chain of thought demonstrations are provided as exemplars in prompting.

Chain-of-Thought Prompting

- Add a series of intermediate reasoning steps in the demonstration examples(s)
- Get the model to explain its reasoning steps before making an answer

Standard Prompting	Chain-of-Thought Prompting
Model Input Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now? A: The answer is 11. Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?	Model Input Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now? A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11. Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?
Model Output A: The answer is 27. ❌	Model Output A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Self-Consistency [Wang et al., 2023]

SELF-CONSISTENCY IMPROVES CHAIN OF THOUGHT REASONING IN LANGUAGE MODELS

Xuezhi Wang^{†‡} Jason Wei[†] Dale Schuurmans[†] Quoc Le[†] Ed H. Chi[†]
Sharan Narang[†] Aakanksha Chowdhery[†] Denny Zhou^{†§}

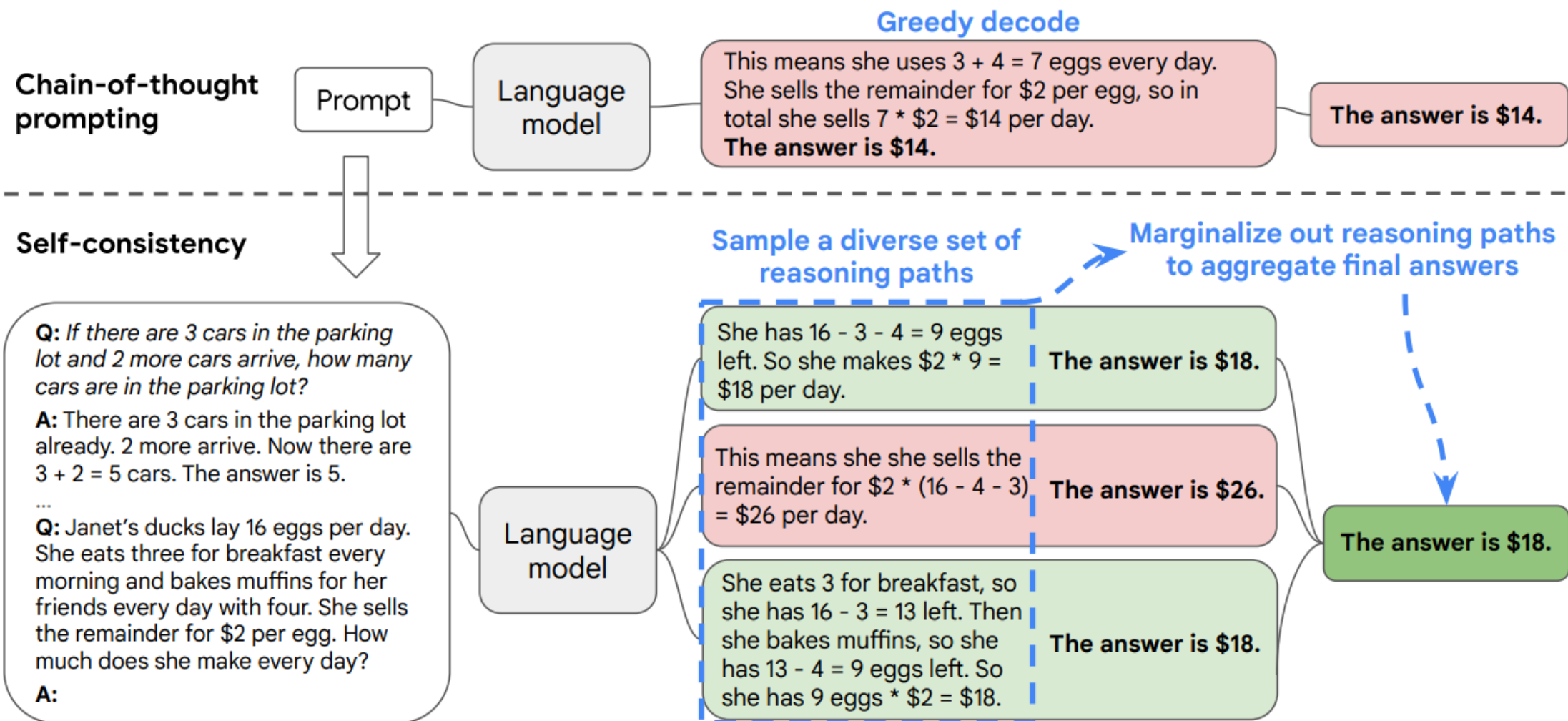
[†]Google Research, Brain Team

[‡]xuezhiw@google.com, [§]dennyzhou@google.com

ABSTRACT

Chain-of-thought prompting combined with pre-trained large language models has achieved encouraging results on complex reasoning tasks. In this paper, we propose a new decoding strategy, *self-consistency*, to replace the naive greedy decoding used in chain-of-thought prompting. It first samples a diverse set of reasoning paths instead of only taking the greedy one, and then selects the most consistent answer by marginalizing out the sampled reasoning paths. Self-consistency leverages the intuition that a complex reasoning problem typically admits multiple different ways of thinking leading to its unique correct answer. Our extensive empirical evaluation shows that self-consistency boosts the performance of chain-of-thought prompting with a striking margin on a range of popular arithmetic and commonsense reasoning benchmarks, including GSM8K (+17.9%), SVAMP (+11.0%), AQuA (+12.2%), StrategyQA (+6.4%) and ARC-challenge (+3.9%).

Self-Consistency



From Few-Shot to Zero-Shot (Instruction Tuning and Alignment)

Our Final Goal

Solve the following Math Word Problem:

John buys 20 cards and 1/4 are uncommon. How many uncommon cards did he get?

0 Math Word Problem training samples

Classify the following paper. Candidate classes are [XXX, XXX, ...]:
Rapid chromatographic technique for preparative separations with ...

0 Paper Classification training samples

Find all the DISEASE entities in the following sentence:
in rats, nitrofurantoin causes pulmonary toxicity

0 Information Extraction training samples

Answer the following question:
Who formulated the zeroth law of thermodynamics?

0 Question Answering training samples



One Model for All Tasks!

5

Organic Chemistry

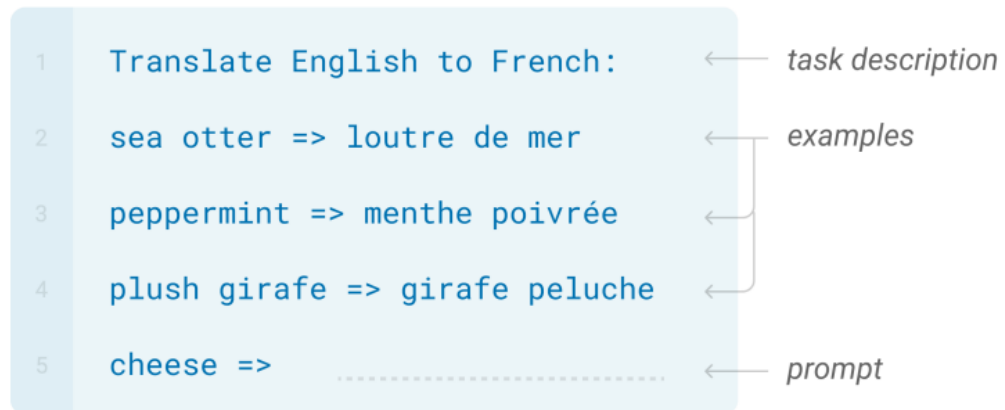
pulmonary toxicity

Ralph H. Fowler

From Few-Shot to Zero-Shot

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



Zero-shot

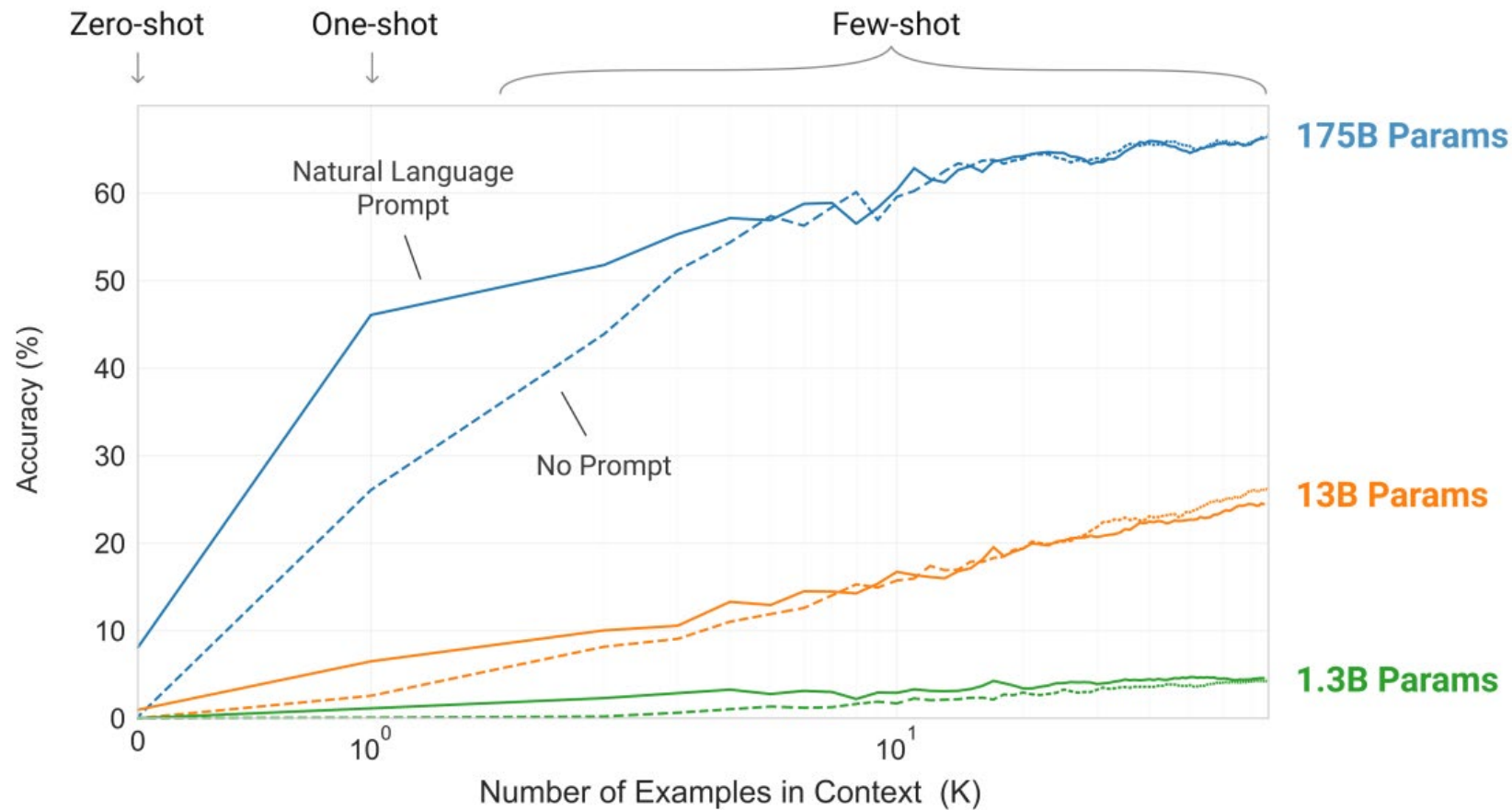
The model predicts the answer given only a natural language description of the task. No gradient updates are performed.



Is GPT-3 a zero-shot learner?

Task Instruction
Only

Task Instruction
+ A Few Examples



Why is the zero-shot setting hard?

- GPT-3 is not good at following an instruction to perform a new task.
 - Because it is never asked to do so during pre-training.
- How to solve this problem?
 - Tune the model to follow task instructions!

$$\sum_i p(\text{ground truth output} \mid \text{instruction, task input})$$

- How can we ensure that the model can handle unseen task instructions when it is actually used?
 - Let it see a sufficiently diverse set of task instructions during training so that it can generalize well
 - What really matters is the number of different tasks, not how many samples each task has
 - Find all available NLP benchmark datasets and convert them into the (instruction, input, output) format
 - Think of lots of creative new tasks and get people to label them

FLAN [Wei et al., 2022] 

FINETUNED LANGUAGE MODELS ARE ZERO-SHOT LEARNERS

Jason Wei*, **Maarten Bosma***, **Vincent Y. Zhao***, **Kelvin Guu***, **Adams Wei Yu**,
Brian Lester, **Nan Du**, **Andrew M. Dai**, and **Quoc V. Le**

Google Research

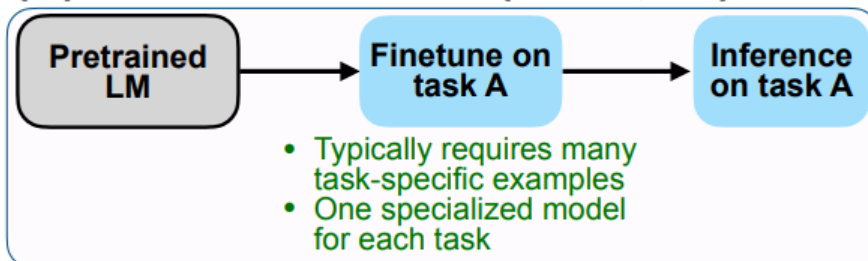
ABSTRACT

This paper explores a simple method for improving the zero-shot learning abilities of language models. We show that *instruction tuning*—finetuning language models on a collection of datasets described via instructions—substantially improves zero-shot performance on unseen tasks.

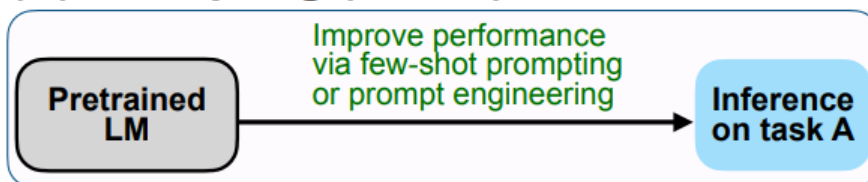
We take a 137B parameter pretrained language model and instruction tune it on over 60 NLP datasets verbalized via natural language instruction templates. We evaluate this instruction-tuned model, which we call FLAN, on unseen task types. FLAN substantially improves the performance of its unmodified counterpart and surpasses zero-shot 175B GPT-3 on 20 of 25 datasets that we evaluate. FLAN even

FLAN

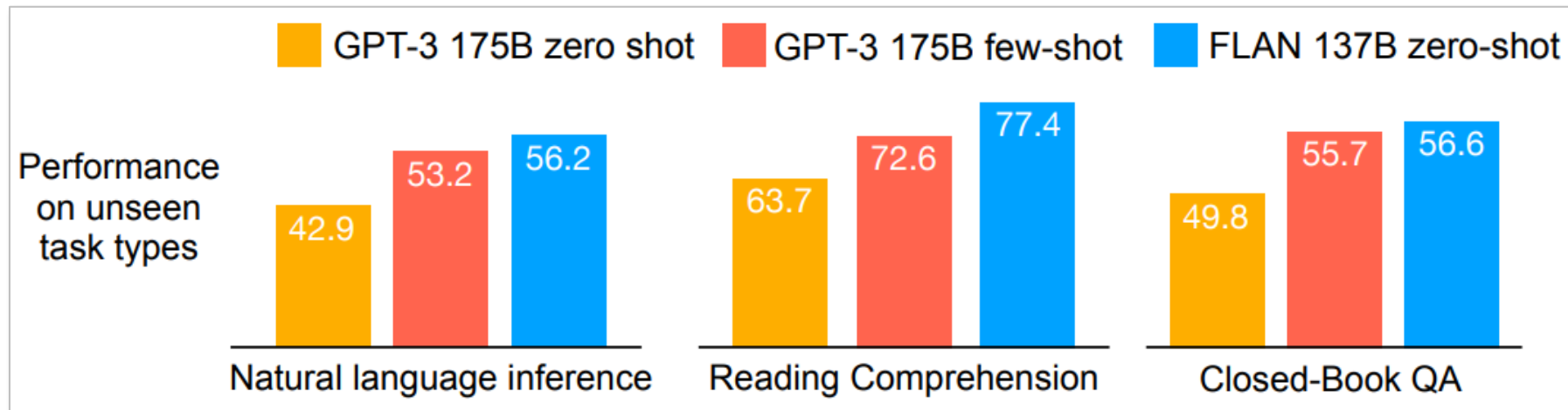
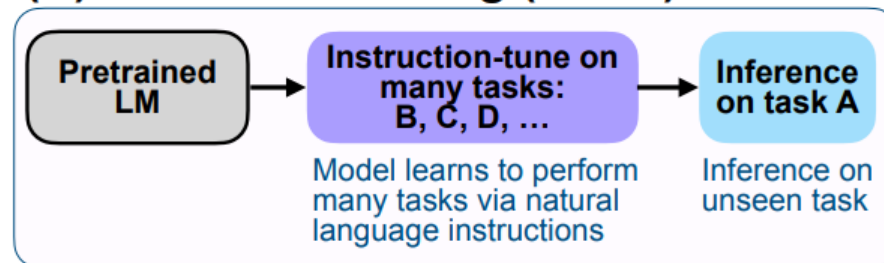
(A) Pretrain–finetune (BERT, T5)



(B) Prompting (GPT-3)



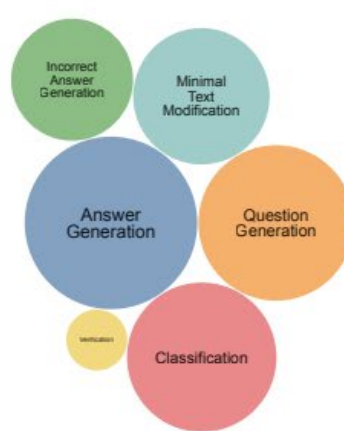
(C) Instruction tuning (FLAN)



Instruction Tuning: A Competition of Data Collection/Annotation



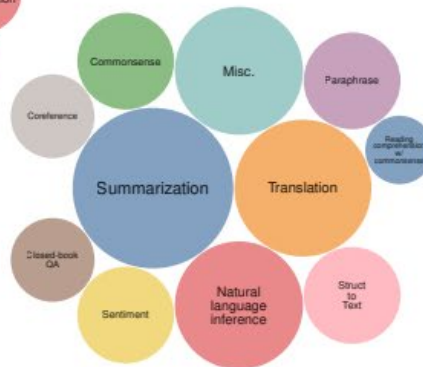
(a) SUP-NATINST (this work)



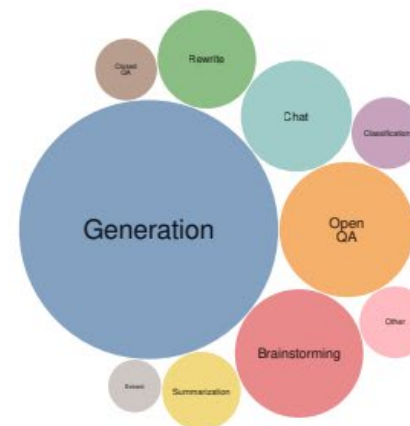
(b) NATINST



(c) PROMPTSOURCE (T0 subset)



(d) FLAN



(e) INSTRUCTGPT

Sup-NatInst: 1,616 tasks

FLAN: 62 tasks

InstructGPT [Ouyang et al., 2022]

Training language models to follow instructions with human feedback

Long Ouyang* **Jeff Wu*** **Xu Jiang*** **Diogo Almeida*** **Carroll L. Wainwright***

Pamela Mishkin* **Chong Zhang** **Sandhini Agarwal** **Katarina Slama** **Alex Ray**

John Schulman **Jacob Hilton** **Fraser Kelton** **Luke Miller** **Maddie Simens**

Amanda Askell[†]

Peter Welinder

Paul Christiano^{*†}

Jan Leike*

Ryan Lowe*

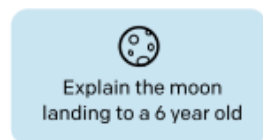
OpenAI

InstructGPT

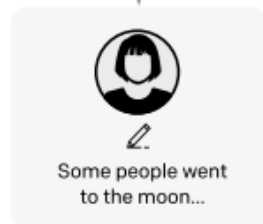
Step 1

Collect demonstration data, and train a supervised policy.

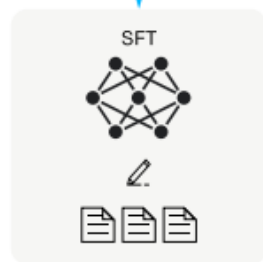
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3 with supervised learning.

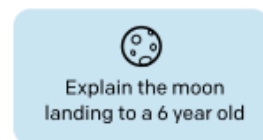


Instruction Tuning

Step 2

Collect comparison data, and train a reward model.

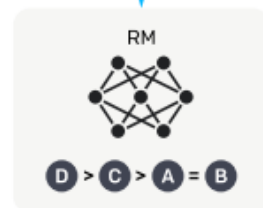
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



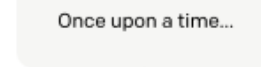
Step 3

Optimize a policy against the reward model using reinforcement learning.

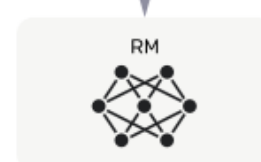
A new prompt is sampled from the dataset.



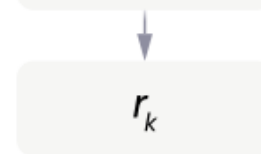
The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.



InstructGPT

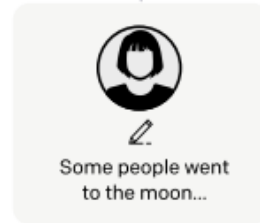
Step 1

Collect demonstration data, and train a supervised policy.

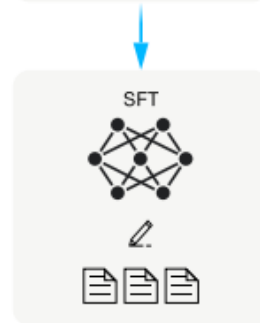
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3 with supervised learning.

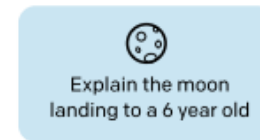


Train another model to imitate human preferences

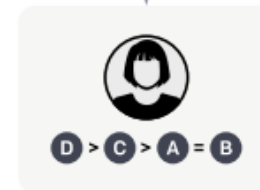
Step 2

Collect comparison data, and train a reward model.

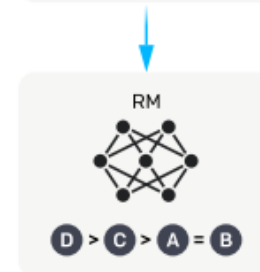
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



Train GPT to satisfy the “human” model

Step 3

Optimize a policy against the reward model using reinforcement learning.

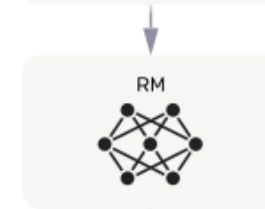
A new prompt is sampled from the dataset.



The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.

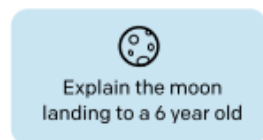


InstructGPT

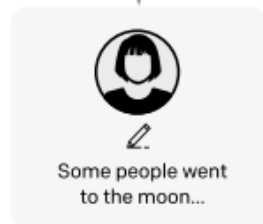
Step 1

Collect demonstration data, and train a supervised policy.

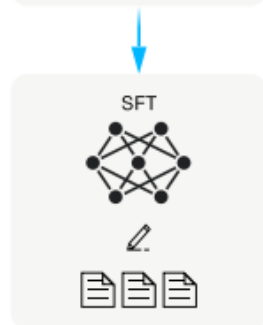
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



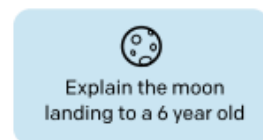
This data is used to fine-tune GPT-3 with supervised learning.



Step 2

Collect comparison data, and train a reward model.

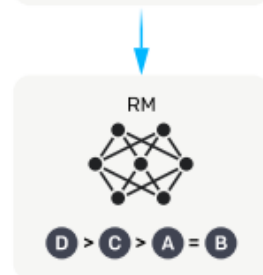
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



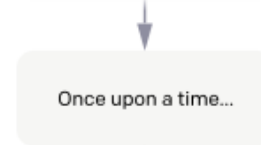
Step 3

Optimize a policy against the reward model using reinforcement learning.

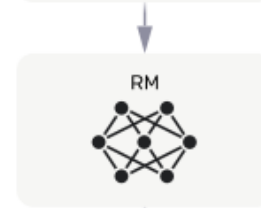
A new prompt is sampled from the dataset.



The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.



Zero-Shot CoT Prompting [Kojima et al., 2022]

Large Language Models are Zero-Shot Reasoners

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Abstract

Pretrained large language models (LLMs) are widely used in many sub-fields of natural language processing (NLP) and generally known as excellent *few-shot* learners with task-specific exemplars. Notably, chain of thought (CoT) prompting, a recent technique for eliciting complex multi-step reasoning through step-by-step answer examples, achieved the state-of-the-art performances in arithmetics and symbolic reasoning, difficult *system-2* tasks that do not follow the standard

“Let’s think step by step.”

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. **X**

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. **✓**

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 **X**

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let’s think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. **✓**

The Power of “*Let’s think step by step.*”

No.	Category	Template	Accuracy
1	instructive	Let’s think step by step.	78.7
2		First, (*1)	77.3
3		Let’s think about this logically.	74.5
4		Let’s solve this problem by splitting it into steps. (*2)	72.2
5		Let’s be realistic and think step by step.	70.8
6		Let’s think like a detective step by step.	70.3
7		Let’s think	57.5
8		Before we dive into the answer,	55.7
9		The answer is after the proof.	45.7
10	misleading	Don’t think. Just feel.	18.8
11		Let’s think step by step but reach an incorrect answer.	18.7
12		Let’s count the number of "a" in the question.	16.7
13		By using the fact that the earth is round,	9.3
14	irrelevant	By the way, I found a good restaurant nearby.	17.5
15		AbraKadabra!	15.5
16		It’s a beautiful day.	13.1
-		(Zero-shot)	17.7

Large Language Models with Search Engines (Retrieval-Augmented Generation)

What if an LLM does not have the knowledge to answer a question?

**You**

What is the most cited paper in WWW 2017? Please list its title and authors.

**ChatGPT**

As of my last update in January 2022, I don't have the real-time citation information, but one of the highly cited papers from WWW 2017 is:

Title: "HIN2Vec: Explore Meta-paths in Heterogeneous Information Networks for Representation Learning"

Authors: Cheng Yang, Zhiyuan Liu, Deli Zhao, Maosong Sun, Edward Y. Chang

Please verify the latest citation counts and updates from reliable sources or academic databases for the most current information.



HIN2Vec was published in **CIKM 2017**.

HIN2Vec was written by **Tao-yang Fu, Wang-Chien Lee, and Zhen Lei**.

ChatGPT 3.5, queried on January 23, 2024

What if an LLM does not have the knowledge to answer a question?

- Why does this happen?
 - **Knowledge cutoffs**: parameters are usually only updated to a particular time
 - **Private data**: data stored in private text or data repositories not suitable for training
 - **Learning failures**: even for data that the model was trained on, it might not be sufficient to get the right answer
- How to solve this?
 - Fine-tune the LLM to know one more fact ❌
 - Teach the LLM to explore the external, up-to-date information ✔️
 - “Give a man *a fish*, and you feed him for a day; teach a man *to fish*, and you feed him for a lifetime.”

Retrieval-Augmented Generation [Lewis et al., 2020]

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Patrick Lewis^{†‡}, Ethan Perez^{*},

Aleksandra Piktus[†], Fabio Petroni[†], Vladimir Karpukhin[†], Naman Goyal[†], Heinrich Küttler[†],

Mike Lewis[†], Wen-tau Yih[†], Tim Rocktäschel^{†‡}, Sebastian Riedel^{†‡}, Douwe Kiela[†]

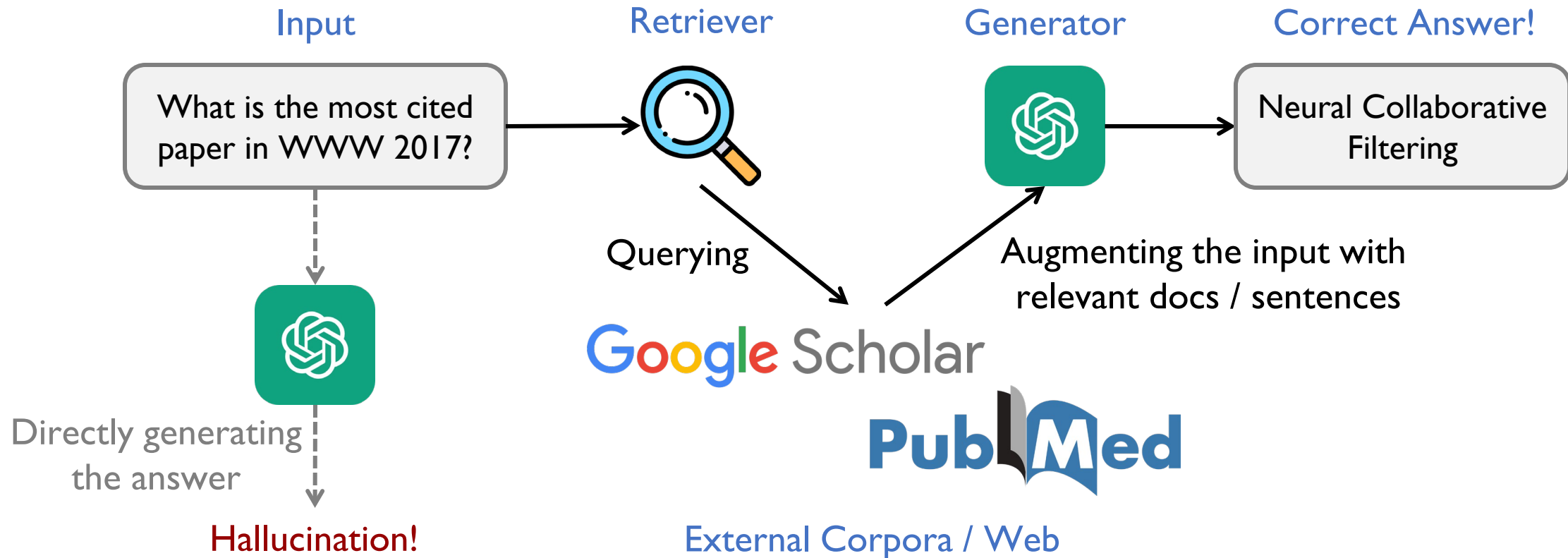
[†]Facebook AI Research; [‡]University College London; ^{*}New York University;
plewis@fb.com

Abstract

Large pre-trained language models have been shown to store factual knowledge in their parameters, and achieve state-of-the-art results when fine-tuned on downstream NLP tasks. However, their ability to access and precisely manipulate

Retrieval-Augmented Generation (RAG)

- Simplest version: single round with a pre-specified plan



What if single-round retrieval is not enough?

What is the most cited paper in WWW 2017? Please list its title and authors.



Searched 5 sites ▾

The most cited paper from WWW 2017 is "Attention is All You Need," which introduced the Transformer model, a groundbreaking architecture in natural language processing that uses attention mechanisms without relying on recurrent or convolutional layers. The authors of the paper are Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin [ARXIV](#) [NEURIPS PROCEEDINGS](#).

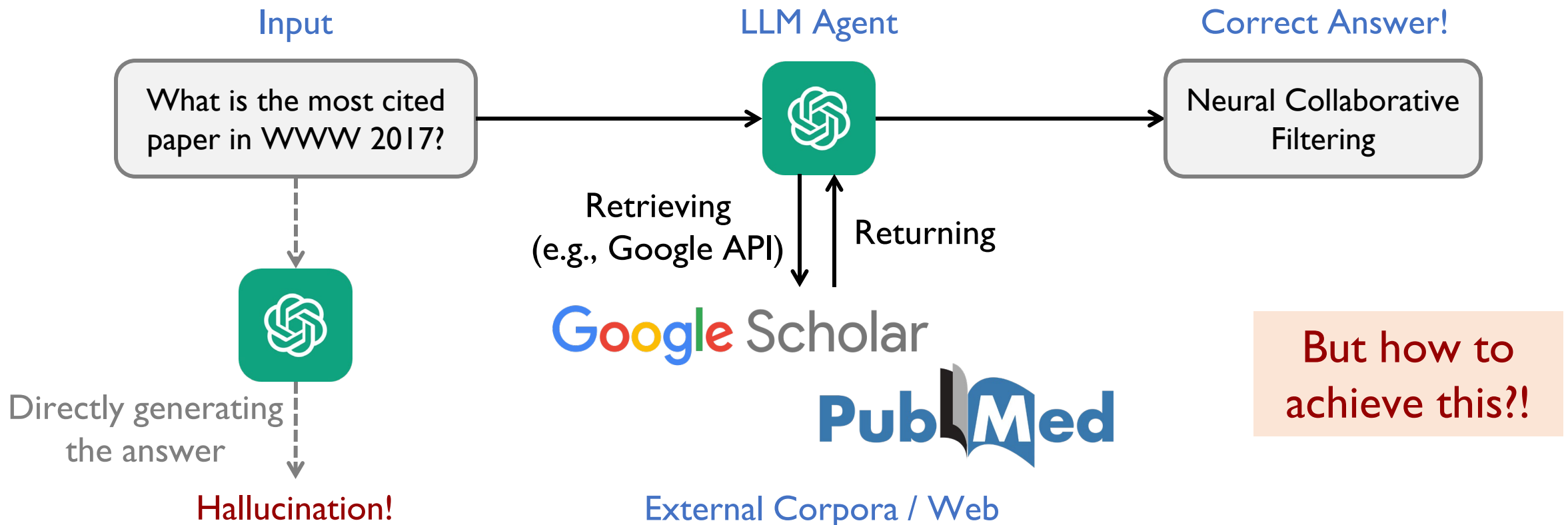


ChatGPT 4o, queried on October 12, 2024

Transformer was published
in **NeurIPS 2017**.

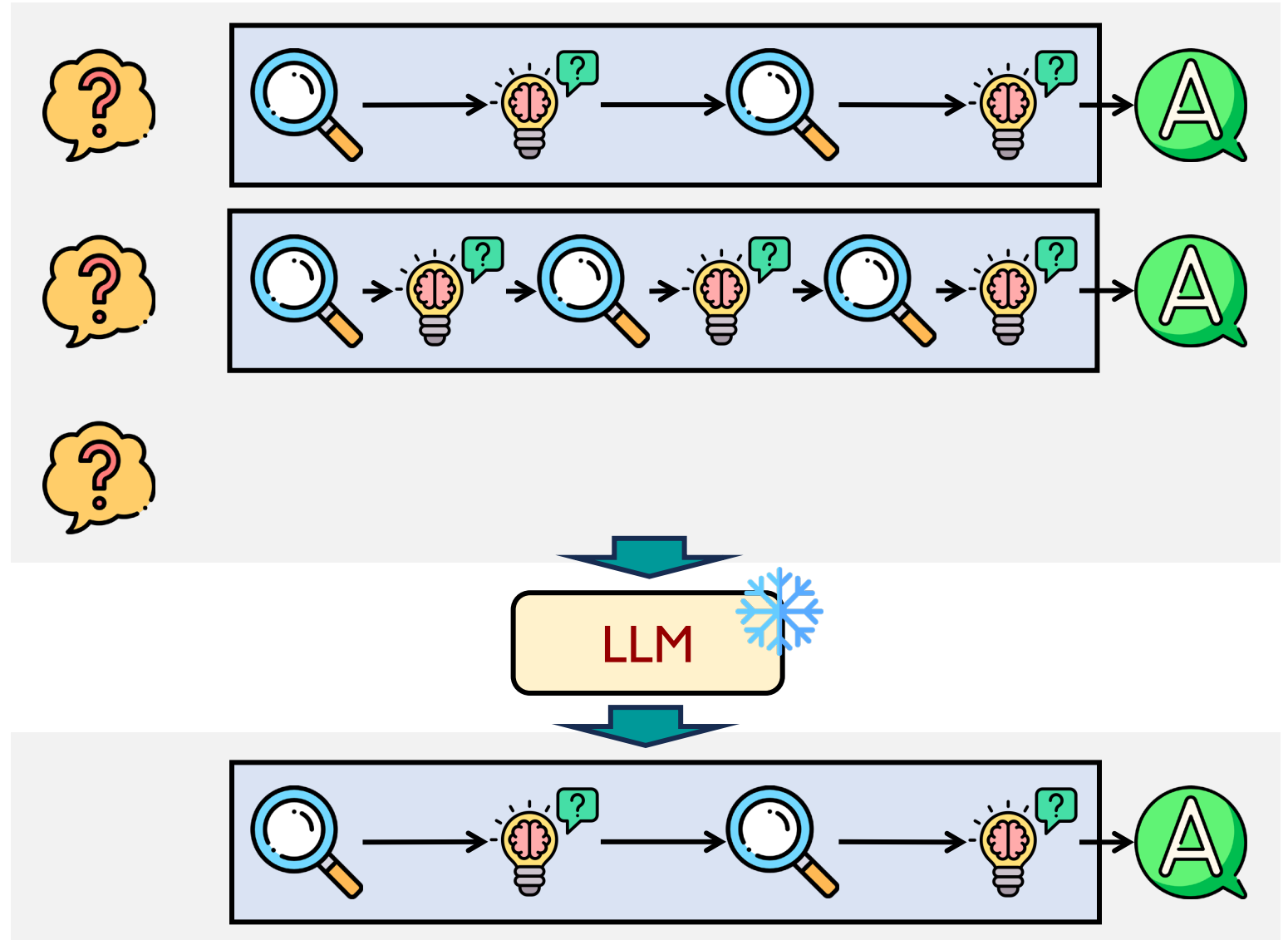
Retrieval-Augmented Generation (RAG)

- **Advanced version:** multi-round, self-adaptive, **agentic**
- Let the model decide when to search, how many rounds to perform, how to formulate queries, ...



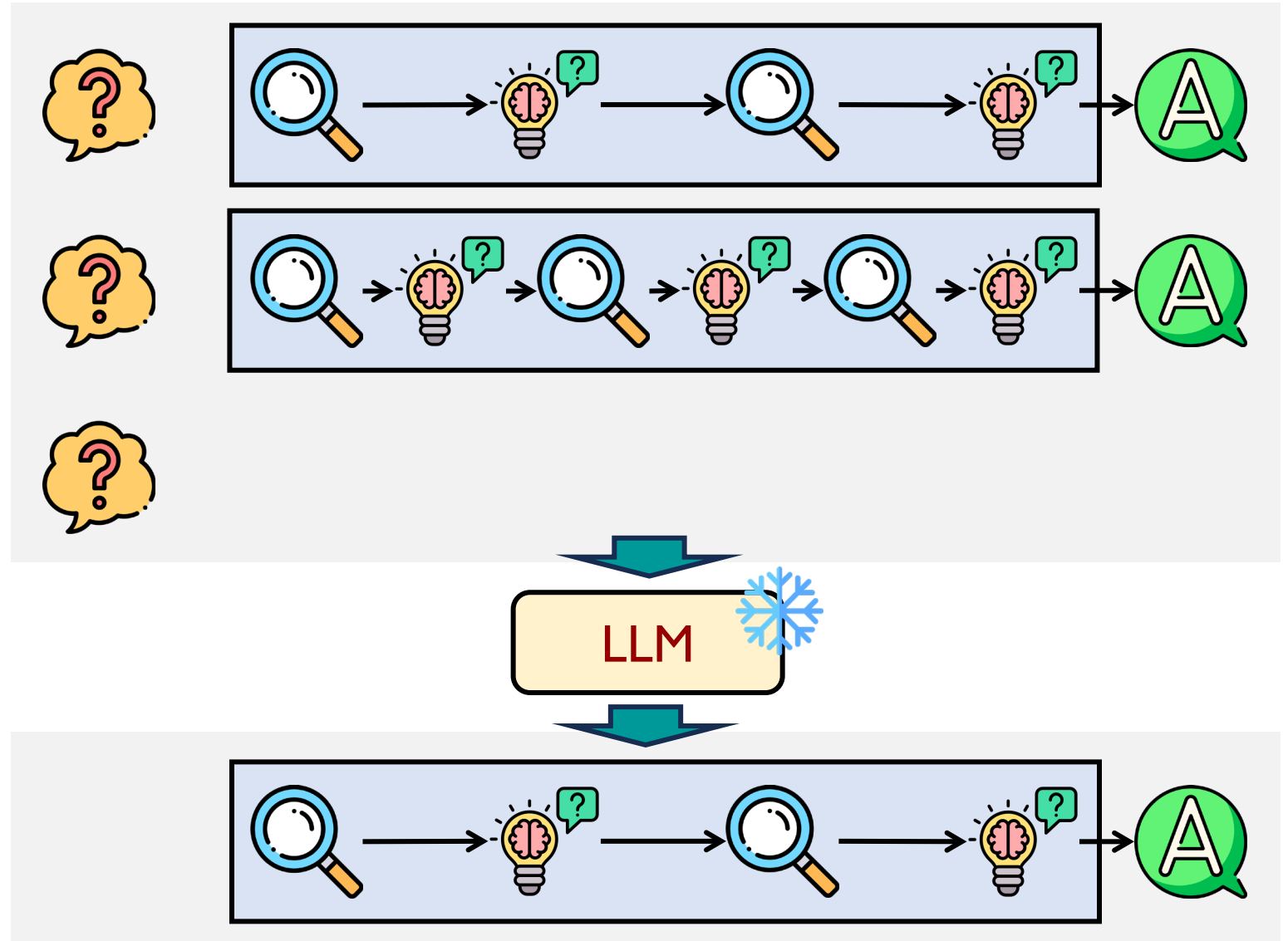
Solution 1: Few-Shot Prompting

- Inference-time technique
- Put reasoning and action examples in the prompt, then ask the question



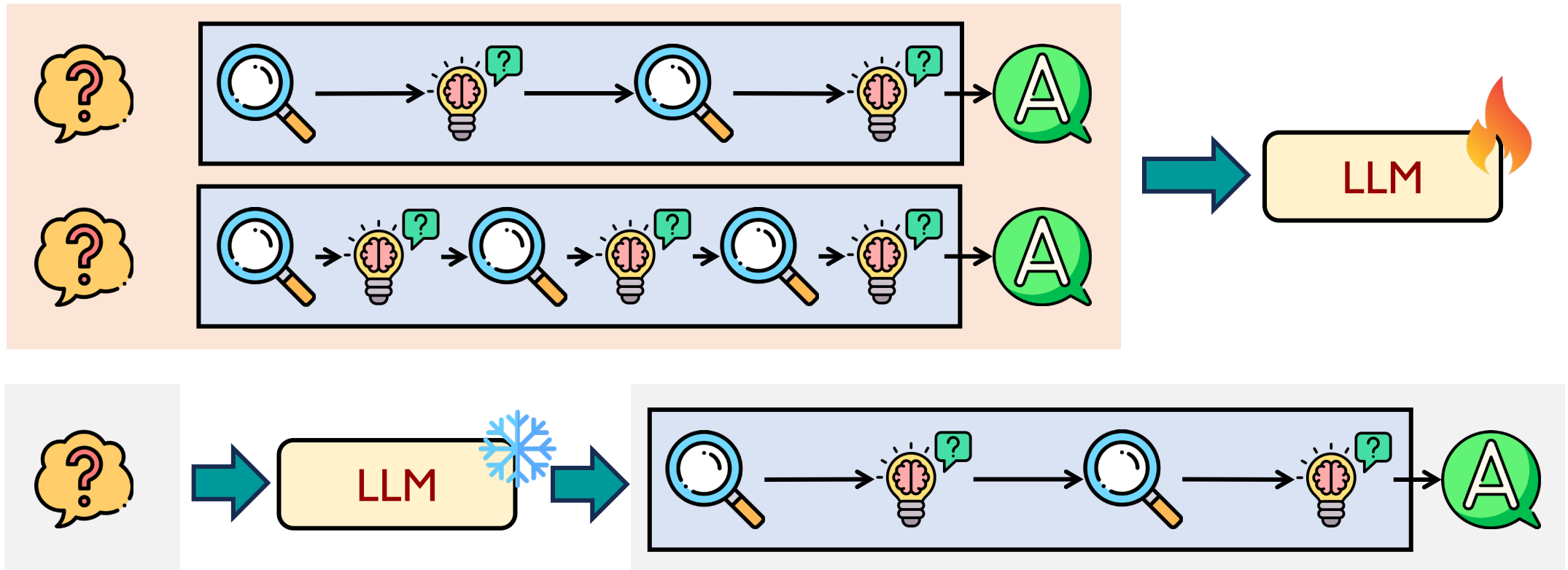
Solution 1: Few-Shot Prompting

- **Limitation:** The ability to call a search engine is not internalized through model parameter updates



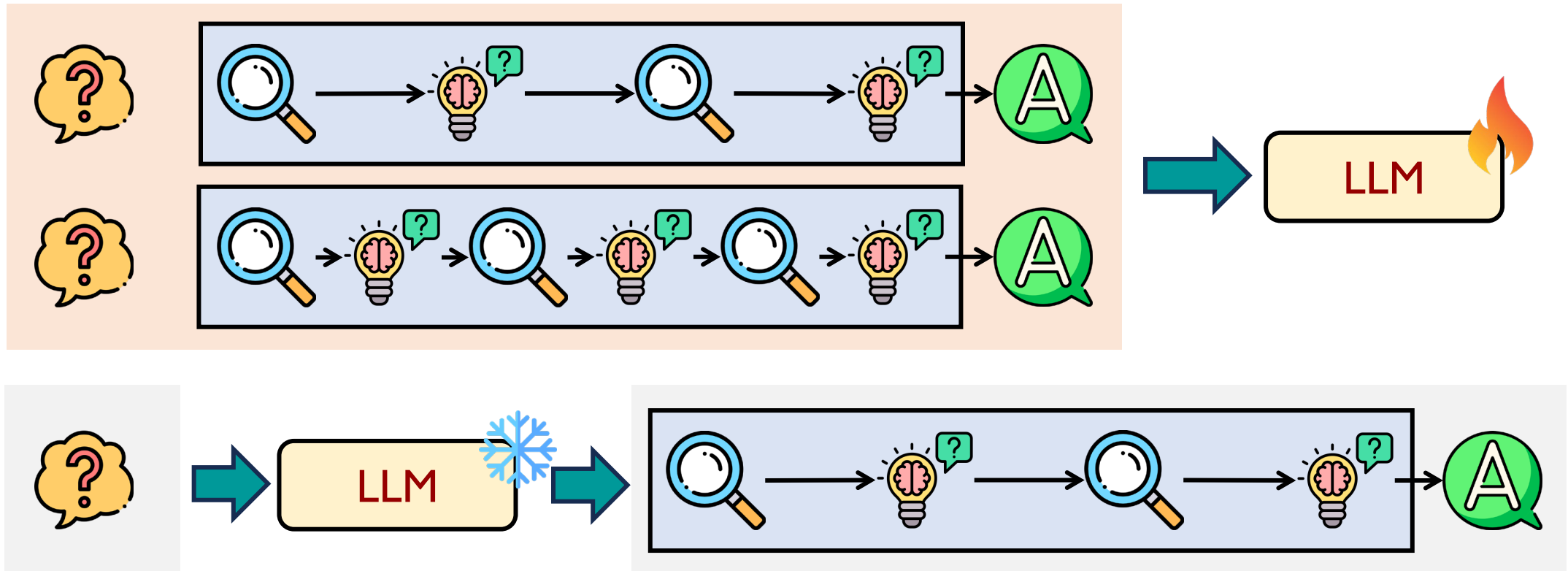
Solution 2: Supervised Fine-Tuning

- Training-based solution
- Collect reasoning and search trajectories to fine-tune the LLM



Solution 2: Supervised Fine-Tuning

- **Limitation 1:** Hard to scale due to reliance on high-quality labeled trajectories
- **Limitation 2:** Poor generalization to unseen search tools (e.g., trained using Google API, but deployed through a search interface we implemented ourselves)



Training LLMs with Reinforcement Learning

DeepSeek-R1: Training LLMs to Reason with RL

- The correct answer (finally...):

What is the most cited paper in WWW 2017? Please list its title and authors.



Reasoning
Started thinking...



The most cited paper from the **WWW 2017** conference is:

Title: *"Neural Collaborative Filtering"*

Authors: Xiangnan He, Lizi Liao, Hanwang Zhang, Liqiang Nie, Xia Hu, and Tat-Seng Chua

DeepSeek-R1 671B, queried on May 8, 2025

Scratchpad of DeepSeek-R1

Reasoning
Started thinking...

Okay, so I need to find out the most cited paper from the WWW 2017 conference. Let me start by recalling that WWW stands for the International World Wide Web Conference, which is a major venue for research related to the web. The 2017 edition would have been the 26th conference.

First, I should figure out how to determine the most cited paper from that specific year. Citation counts can be tracked through databases like Google Scholar, Semantic Scholar, or the ACM Digital Library since ACM often sponsors these conferences. However, citation counts can vary over time, so the most cited paper might change as newer papers accumulate citations. But since the user is asking about the current most cited, I need up-to-date information.

I'll start by checking the proceedings of WWW 2017. The conference usually publishes its proceedings, and sometimes the awards or notable papers are mentioned. If I can find the list of accepted papers, I can then look up each of them on Google Scholar to check their citation counts. But that's time-consuming. Alternatively, maybe there's a known paper from that year that's highly cited. For example, in recent years, transformer-based models like BERT or attention mechanisms have been highly cited, but 2017 might be a bit early for those.

Wait, 2017 is the year when the "Attention Is All You Need" paper was published, but that was at NIPS, not WWW. So maybe that's not relevant here. Let me think. In the realm of web conferences, topics like graph neural networks, recommendation systems, network analysis, and so on are common.

Another approach is to search for "most cited papers from WWW 2017" on Google or academic search engines. Let me try that. Searching for that exact phrase, I find some resources. For instance, there's a page on the WWW 2017 conference website that lists accepted papers, but I need to check if they have any awards or most cited mentions. Alternatively, Semantic Scholar might have a list sorted by citations.

Let me check Semantic Scholar. Going to their website, I can search for papers from the WWW 2017 conference. Applying the filters for the year 2017 and the venue WWW, then sorting by citations. Let's see. The top result there might be the one. Alternatively, Google Scholar can be used by searching for the conference name and year, then sorting by citations.

Upon doing a quick search, I find that the paper "Inductive Representation Learning on Large Graphs" by William L. Hamilton, Rex Ying, and Jure Leskovec is highly cited. Wait, but that was published at NIPS 2017, right? Wait, no, maybe I'm confused. Let me check the details. The Hamilton et al. paper on GraphSAGE was indeed presented at NIPS 2017. So maybe that's not the one.

Alternatively, another paper from WWW 2017 that's highly cited could be related to graph convolutional networks or something similar. Let me check the proceedings. For example, "Graph Convolutional Matrix Completion" by Rianne van den Berg, Thomas N. Kipf, and Max Welling was published in 2017, but I think that's at a different conference.

Another possibility is "Neural Collaborative Filtering" by Xiangnan He, Lizi Liao, Hanwang Zhang, Liqiang Nie, Xia Hu, and Tat-Seng Chua. Wait, but that paper was published in WWW 2017? Let me confirm. According to Google Scholar, that paper was indeed published in the Proceedings of the 26th International Conference on World Wide Web in 2017. It's possible that this paper is highly cited as it introduces neural networks into collaborative filtering, a popular approach in recommendations.

Checking the citation counts: As of now, "Neural Collaborative Filtering" has over 5,000 citations on Google Scholar, which is very high. Another paper, "DeepFM: A Factorization-Machine based Neural Network for CTR Prediction" has over 2,000 citations. There might be others. Wait, but I need to confirm which one has the highest.

Alternatively, "Inductive Representation Learning on Large Graphs" (GraphSAGE) by Hamilton et al. has over 6,000 citations, but that's from NIPS 2017, not WWW. So perhaps the most cited from WWW 2017 is "Neural Collaborative Filtering". Let me cross-verify.

Looking at the WWW 2017 accepted papers list, "Neural Collaborative Filtering" is indeed there. The authors are Xiangnan He, Lizi Liao, Hanwang Zhang, Liqiang Nie, Xia Hu, and Tat-Seng Chua. That paper is foundational in combining neural networks with collaborative filtering, which explains its high citation count.

Alternatively, another paper from WWW 2017 that's highly cited is "StarSpace: Embed All The Things!" by Ledell Wu, Adam Fisch, Sumit Chopra, Keith Adams, Antoine Bordes, and Jason Weston. This paper also has a substantial number of citations, but I think it's less than "Neural Collaborative Filtering".

To be sure, checking Google Scholar for "Neural Collaborative Filtering" shows over 5,000 citations as of 2023. "DeepFM" has around 2,000. Another paper, "Metapath2vec: Scalable Representation Learning for Heterogeneous Networks" by Yuxiao Dong, Nitesh V. Chawla, and Ananthram Swami, which was also presented at WWW 2017, has over 1,500 citations.

Therefore, it's likely that "Neural Collaborative Filtering" by He et al. is the most cited paper from WWW 2017. However, I should also check if there's any other paper with higher citations. For example, "Attention Is All You Need" is from 2017 but not WWW. "Dynamic Routing Between Capsules" by Hinton et al. is also from 2017 but a different conference.

Another possible candidate is "Semi-Supervised Classification with Graph Convolutional Networks" by Kipf and Welling, but that was published at ICLR 2017, so not WWW.

Therefore, after checking the citation counts and cross-referencing with the conference proceedings, "Neural Collaborative Filtering" by Xiangnan He and colleagues seems to be the most cited paper from WWW 2017. However, I should verify once more. Let me check another source.

On Semantic Scholar, searching for WWW 2017 papers sorted by citations: the top result is "Neural Collaborative Filtering" with 5,442 citations as of October 2023. The next is "Metapath2vec" with 2,035 citations. Then "DeepFM" with 1,821. So yes, "Neural Collaborative Filtering" is the most cited.

Thus, the answer should be that paper.

Scratchpad of DeepSeek-R1

Another approach is to search for "most cited papers from WWW 2017" on Google or academic search engines. Let me try that. Searching for that exact phrase, I find some resources. For instance, there's a page on the WWW 2017 conference website that lists accepted papers, but I need to check if they have any awards or most cited mentions. Alternatively, Semantic Scholar might have a list sorted by citations.

Let me check Semantic Scholar. Going to their website, I can search for papers from the WWW 2017 conference. Applying the filters for the year 2017 and the venue WWW, then sorting by citations. Let's see. The top result there might be the one. Alternatively, Google Scholar can be used by searching for the conference name and year, then sorting by citations.

Checking the citation counts: As of now, "Neural Collaborative Filtering" has over 5,000 citations on Google Scholar, which is very high. Another paper, "DeepFM: A Factorization-Machine based Neural Network for CTR Prediction" has over 2,000 citations. There might be others. Wait, but I need to confirm which one has the highest.

Alternatively, "Inductive Representation Learning on Large Graphs" (GraphSAGE) by Hamilton et al. has over 6,000 citations, but that's from NIPS 2017, not WWW. So perhaps the most cited from WWW 2017 is "Neural Collaborative Filtering". Let me cross-verify.

DeepSeek-R1 [Guo et al., 2025]

Article | [Open access](#) | Published: 17 September 2025

DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning

[Daya Guo](#), [Dejian Yang](#), [Haowei Zhang](#), [Junxiao Song](#), [Peiyi Wang](#), [Qihao Zhu](#), [Runxin Xu](#), [Ruoyu Zhang](#), [Shirong Ma](#), [Xiao Bi](#), [Xiaokang Zhang](#), [Xingkai Yu](#), [Yu Wu](#), [Z. F. Wu](#), [Zhibin Gou](#), [Zhihong Shao](#), [Zhuoshu Li](#), [Ziyi Gao](#), [Aixin Liu](#), [Bing Xue](#), [Bingxuan Wang](#), [Bochao Wu](#), [Bei Feng](#), [Chengda Lu](#), ... [Zhen Zhang](#) [+ Show authors](#)

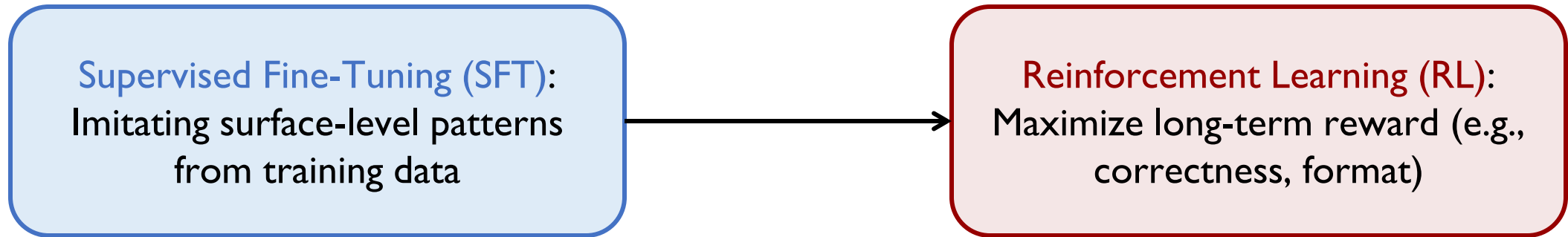
[Nature](#) **645**, 633–638 (2025) | [Cite this article](#)

278k Accesses | **53** Citations | **789** Altmetric | [Metrics](#)

Abstract

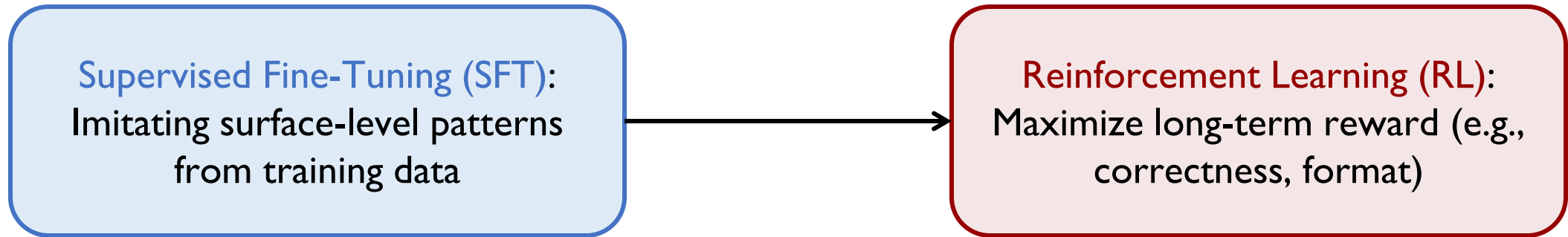
General reasoning represents a long-standing and formidable challenge in artificial intelligence (AI). Recent breakthroughs, exemplified by large language models (LLMs)^{1,2} and

DeepSeek-R1: Training LLMs to Reason with RL



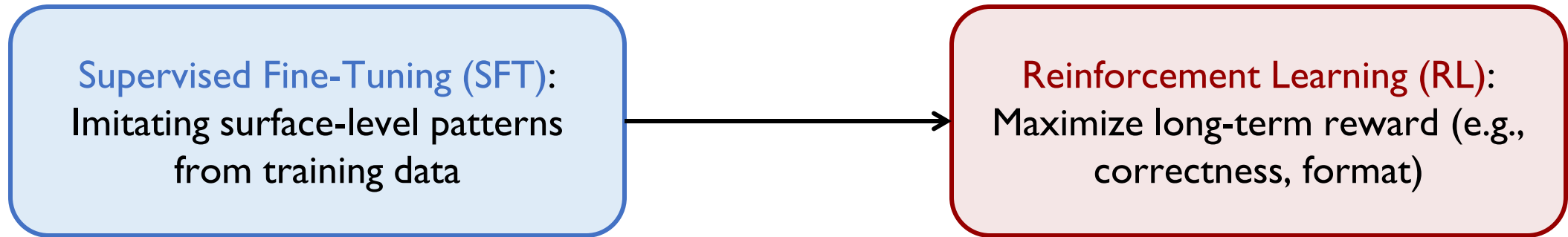
- **Supervised Fine-Tuning (SFT):** Given a question, maximize the likelihood that the LLM generates the reference answer.
 - If the reference answer is 0.5, then generating 1/2 is wrong.
 - If the LLM is asked to write code for a specific function, only an exact match with the reference answer is considered fully correct. A program that differs from the reference by just one token (even if it does not run at all) is regarded as better than an alternative implementation that correctly achieves the same functionality.

DeepSeek-R1: Training LLMs to Reason with RL



- **Reinforcement Learning with Verifiable Rewards (RLVR):** When the LLM does something correct according to our rubrics, it gets certain reward. The model is trained to maximize the reward.
 - **Format Reward:**
 - Has `<think>` and `</think>` tokens in its output; puts its intermediate reasoning steps between these two tokens.
 - Has `<answer>` and `</answer>` tokens in its output; puts its final answer between these tokens
 - ...

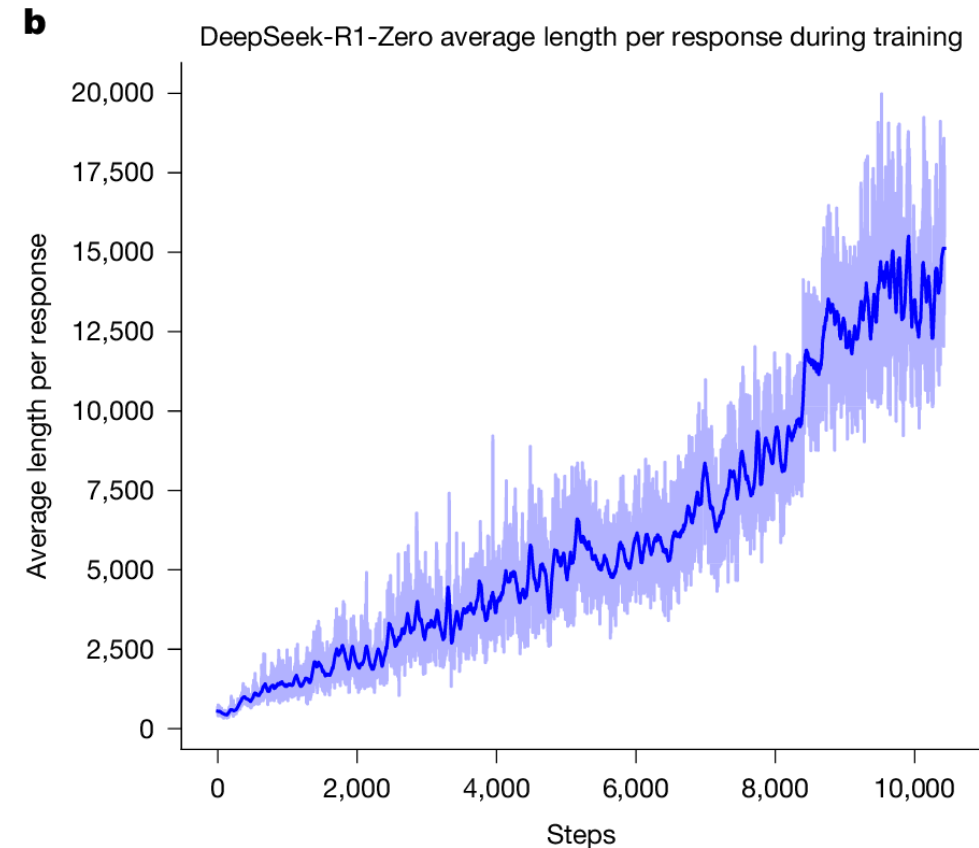
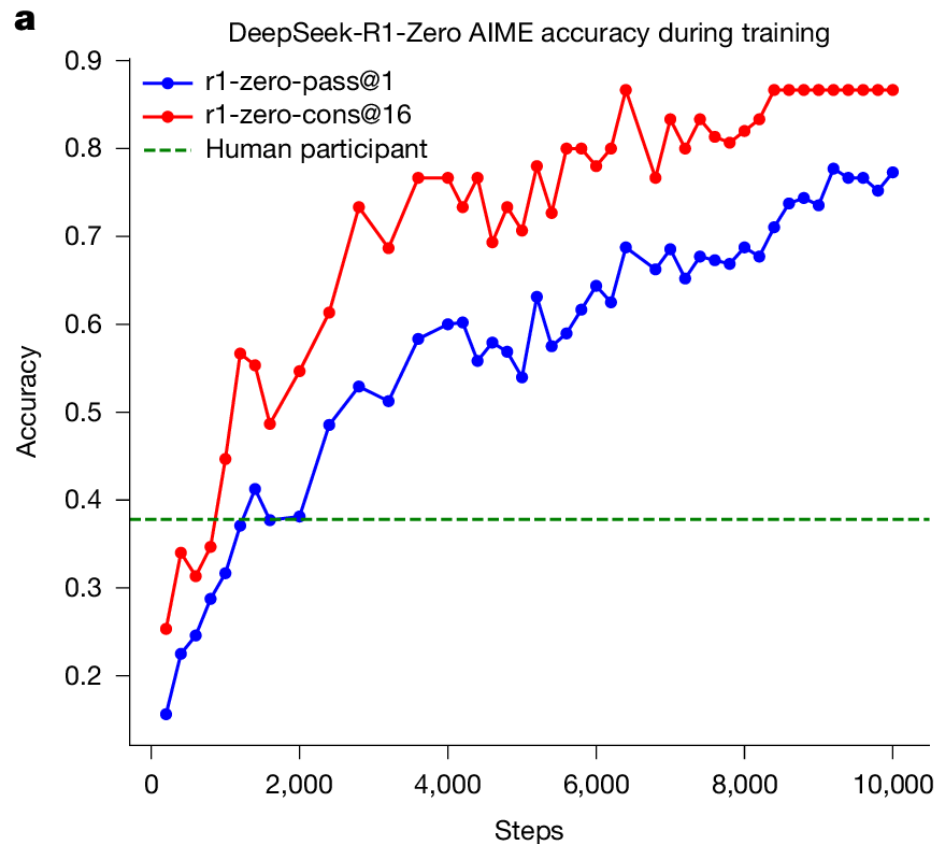
DeepSeek-R1: Training LLMs to Reason with RL



- **Reinforcement Learning with Verifiable Rewards (RLVR):** When the LLM does something correct according to our rubrics, it gets certain reward. The model is trained to maximize the reward.
 - **Correctness Reward:**
 - The answer (i.e., tokens between `<answer>` and `</answer>`) should be “equivalent” to the reference answer (according to a verifier).
 - If the model does not generate the correct answer, it will **NOT** know the answer after this iteration; it will only know that the reward is low.
 - This question can be used repeatedly until the model generates a good answer.

DeepSeek-R1: Training LLMs to Reason with RL

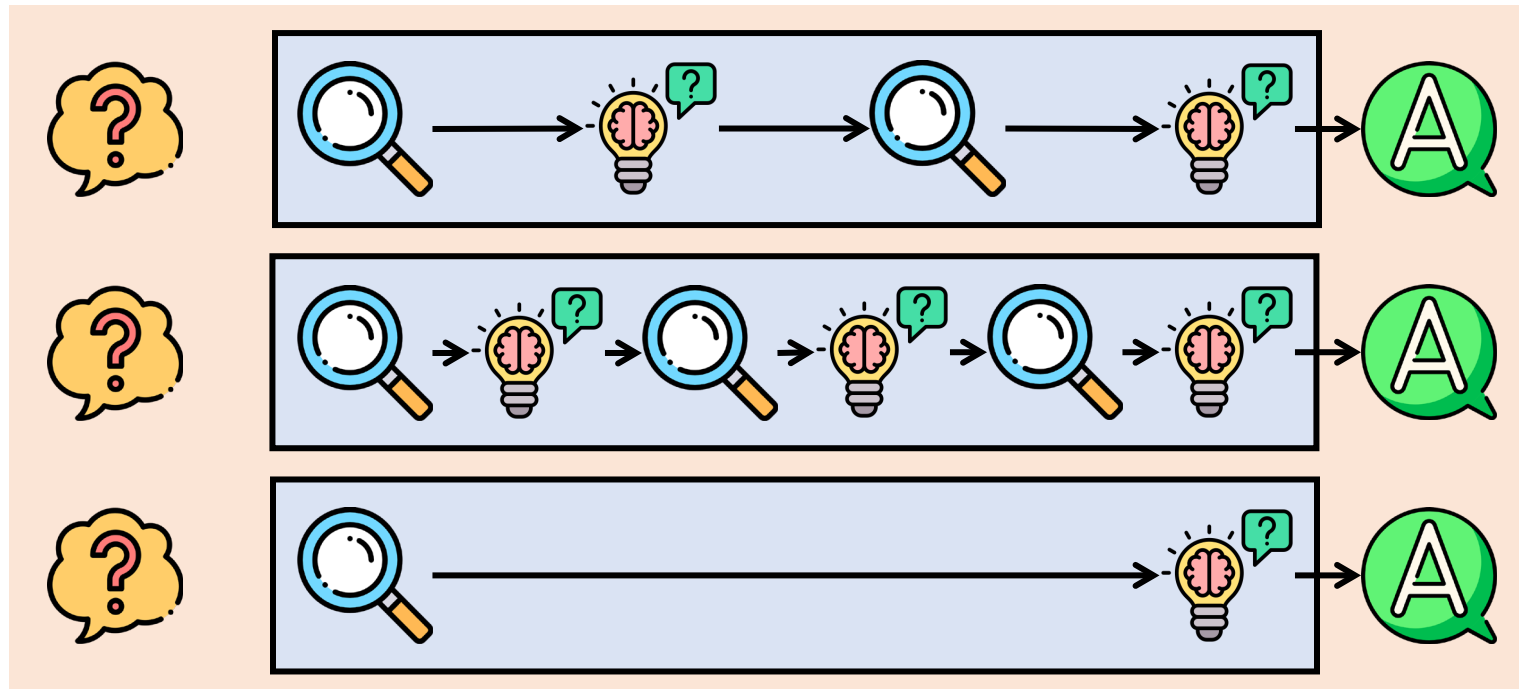
- When the model can repeatedly try while only receiving binary feedback (correct/incorrect), it is forced to engage in longer reasoning before reaching the correct answer.
- Experiments show this is effective.



Training LLMs to Search with RL?

- Motivation:
 - Because DeepSeek-RL is popular ✗
 - Because we no longer need annotated trajectories! ✓

I do not have this, ...



..., but I have this
(which is enough for RL)!



Search-R1 [Jin et al., COLM 2025]

Search-R1: Training LLMs to Reason and Leverage Search Engines with Reinforcement Learning

Bowen Jin¹, Hansi Zeng², Zhenrui Yue¹, Jinsung Yoon³, Sercan Ö. Arık³, Dong Wang¹,
Hamed Zamani², Jiawei Han¹

¹ Department of Computer Science, University of Illinois at Urbana-Champaign

² Center for Intelligent Information Retrieval, University of Massachusetts Amherst

³ Google Cloud AI Research

{bowenj4,zhenrui3,dwang24,hanj}@illinois.edu, {hzeng, zamani}@cs.umass.edu
{jinsungyoon,soarik}@google.com

Abstract

Efficiently acquiring external knowledge and up-to-date information is essential for effective reasoning and text generation in large language models (LLMs). Prompting advanced LLMs with reasoning capabili-



Thank You!